Model Evaluation Plan: Evaluating ASR Robustness Under Overlapping English Speech

Jamin

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Contents

1 Introduction **3**

2 Literature Review **5**
2.1 Foundational Knowledge . . . 5
2.1.1 Multi-Speaker Corpora for ASR Research . . . 5
2.1.2 Overlap Modelling and Evaluation Metrics for Multi-Speaker ASR . . . 6
2.1.3 ASR Architectures and Benchmark Systems for Robust Speech Transcription 6
2.2 Establishing a Gap in the Literature . . . 7
2.3 Recent Advances . . . 7
2.3.1 Realism and Generalisation in Overlap Modelling . . . 7
2.3.2 Integrated Pipelines and End-to-End ASR for Overlapping Speech . . . 7
2.3.3 Unified Metrics for Multi-Speaker ASR Evaluation . . . 8
2.3.4 Summary . . . 9

3 Architectural Design **10**
3.1 Design Objective and Principles . . . 10
3.2 System Overview . . . 10
3.2.1 System Boundary and Design Rationale . . . 10
3.3 Artifact Structure . . . 11
3.3.1 Synthetic Audio Manifest Structure . . . 11
3.3.2 Transcription Output Schema . . . 11
3.3.3 Metric Output Schema . . . 11
3.4 Synthetic Mixture Generation . . . 12
3.4.1 Speaker and Clip Selection . . . 13
3.4.2 Offset Scheduling . . . 13
3.4.3 Noise Addition . . . 14
3.5 ASR Inference and Experiment Orchestration . . . 14
3.5.1 Fairness Controls . . . 15
3.6 Evaluation Architecture . . . 15
3.6.1 Established Metric Modules . . . 15
3.6.2 Proposed Metric Module: Double-Stream Serialised WER (DSS-WER) . 16
3.6.3 Beam-Search Approximation for DSS-WER . . . 16
3.6.4 Metric Output Schema . . . 19
3.7 Benchmarking . . . 19
3.8 Real-Data Segmentation and Evaluation Pipeline . . . 20
3.9 Statistical Analysis Module . . . 20
3.10 Design Trade-offs and Limitations . . . 20
3.11 Reproducibility and Experimental Setup . . . 20

4 Experimental Design **26**
4.1 Objective and Scope . . . 26
4.2 Research Questions . . . 26
4.3 Design Principles . . . 27
4.4 Track 1: Controlled Synthetic Benchmark . . . 27
4.4.1 Speech and Noise Sources . . . 27
4.4.2 Synthetic Mixture Generation . . . 28
4.4.3 Noise Addition and SNR Control . . . 28
4.4.4 Experimental Conditions and Corpus Size . . . 28
4.5 Track 2: Real-world Data Transferability Evaluation . . . 29
4.6 ASR Models Under Evaluation . . . 30
4.7 Scoring Metrics . . . 30
4.7.1 Reference and Hypothesis Structure . . . 30
4.7.2 Metrics . . . 31
4.7.3 Metric Application Rules . . . 31
4.8 Experimental Unit and Comparison Strategy . . . 31
4.9 Track 3: Hypothesis Testing of DSS-WER . . . 31
4.9.1 Primary Hypothesis . . . 32
4.9.2 Condition-Specific Hypotheses . . . 32
4.9.3 Test Procedure . . . 32
4.9.4 Interpretation . . . 33
4.10 Reproducibility . . . 33
4.11 Scope and Limitations . . . 33

5 Statistical Analysis **34**
5.1 Analytical Aims and Data Provenance . . . 34
5.2 Variable Selection . . . 35
5.2.1 Dependent Variables . . . 35
5.2.2 Explanatory Variables . . . 35
5.3 Statistical Methods . . . 36
5.4 Overview of the Evaluation Results . . . 36
5.4.1 Scope 1: Cross-Model Comparison . . . 36
5.4.2 Inference Time . . . 36
5.5 Scope 2: Full Condition Analysis of wav2vec-2.0 . . . 37
5.6 RQ1: Effects of Overlap Ratio and Signal-to-Noise Ratio . . . 39
5.6.1 Effect of Overlap Ratio . . . 39
5.6.2 Effect of Signal-to-Noise Ratio . . . 40
5.6.3 Summary . . . 41
5.7 RQ2: Comparison of ASR Models . . . 41
5.7.1 Overall Comparison . . . 41
5.7.2 Extreme Cases . . . 42
5.7.3 Inference Efficiency . . . 44
5.7.4 Summary . . . 44
5.8 RQ3: Transferability of ASR Performance Trends . . . 44
5.9 RQ4: Evaluation of DSS-WER . . . 45
5.10 Summary of Findings . . . 47

A Pseudocode for DSS-WER Beam Search **52**

B Software Configuration **53**
B.1 Software Dependencies . . . 53
B.2 ASR Models and Configurations . . . 53

Chapter 1

Introduction

Automatic speech recognition (ASR) has improved substantially in recent years due to advances in deep learning, large-scale training data, and computational resources. However, performance remains much less reliable in realistic conversational settings, particularly when multiple speakers overlap, and background noise is present. To develop and evaluate models under these conditions, robust, realistic datasets are needed. The AMI meeting corpus [1] and the CHIME-6 dataset [2] provide important examples of real-world multi-speaker datasets with realistic, detailed annotations.

Two major factors that degrade ASR performance are overlapping speech and environmental noise. Both are common in natural human conversation [3, 4]. Overlapping speech is the time interval during which two or more speakers speak simultaneously. At the same time, background noise is any unintended sound that does not contribute to the conversation. Overlapping speech violates the single-active-speaker assumption underlying many conventional ASR systems. Natural conversations often involve single-speaker segments and partially overlapping regions [3], where standard metrics such as word error rate (WER) might fail to reflect the accuracy [5, 6]. AS a result, overlap-aware and speaker-aware measures such as concatenated minimum-permutation WER (cpWER), Optimal Reference Combination WER (ORC-WER), and diarisation invariant cpWER (DI-cpWER) are proposed [7].

Due to the difficulties and cost of collecting and annotating real-world multi-speaker datasets, synthetic mixtures have emerged as an important alternative. Existing datasets have improved realism by adding ambient noise, reverberation and increasing the diversity of recording conditions and scenarios. However, most were fixed rather than flexible for downstream training to systematically produce more data and varying overlap, noise, duration and speaker count.

Despite this progress, a practical gap remains. Real meeting conversations preserve natural patterns, but this same naturalness poses lexical and acoustic challenges, limiting control over the degradation factors during training and evaluation. On the other hand, existing synthetic datasets are easier to manipulate, but they are typically built with fixed or partially controlled parameters. As a result, it remains difficult to isolate how factors such as overlap ratio, signal-to-noise ratio, mixture duration and speaker count individually and jointly affect ASR performance.

This dissertation addresses the gap by introducing a synthetic mixture generation framework that takes clean speaker recordings, transcription and noise recording as input and outputs mixtures with a user-specified overlap ratio, signal-to-noise ratio, audio length and speaker count, along with timestamped transcripts. Utilising this framework, this study investigates three main questions: how ASR performance changes with overlap and noise, how transferable is the trend observed in synthetic data to real-world, and how publicly available ASR models compare under different conditions.

This framework provides more transparency and information in model evaluation by separating different conditions by degradation factors. In the case where robustness changes with degradation conditions, single-point evaluation becomes underrepresentative. Hence, instead

of simply reporting under a single dataset, this framework provides an opportunity to analyse how different models behave in a range of different conditions, identifying the strengths and weaknesses of different models.

The structure of the dissertation is as follows: Chapter 2 reviews the related work on ASR robustness, multi-speaker datasets and evaluation metrics. Chapter 3 describes architectural design choices used in the experiments. Chapter 4 details the experimental design, including the generation of synthetic mixtures and the evaluation on real-world data. Chapter 5 outlines and reports the results of experiments. Chapter 6 discusses the implications of the findings, limitations of the study and directions for future work. Finally, Chapter 7 concludes the dissertation by summarising the key contributions and insights gained from the research.

Chapter 2

Literature Review

This chapter reviews the literature relevant to evaluating the robustness of automatic speech recognition (ASR) to overlapping speech. It is structured into three sections. The first section summarises established knowledge about multi-speaker datasets, synthetic mixture generation, overlap modelling, evaluation metrics and ASR architectures. The second section identifies the primary gaps motivating this dissertation. That includes the fact that existing resources do not jointly provide realistic overlap, factor-level control, and fair evaluation when comparing references and hypotheses with different structures. The third section explores recent work that has attempted to address these gaps through more realistic synthesis, more advanced pipelines and end-to-end ASR.

2.1 Foundational Knowledge

2.1.1 Multi-Speaker Corpora for ASR Research

Audio transcription research has depended on a relatively small set of corpora that preserve the acoustic and interactional realism of natural conversations. The *AMI Meeting Corpus* is one of the foundational resources for multi-speaker ASR because it provides approximately 100 hours of monitored meetings with rich annotations and a variety of naturally occurring and scenario-driven interactions [8]. The *CHiME-6* challenge further extended that problem into distant multi-microphone, unsegmented recordings in home environments, introducing a benchmark where diarisation and recognition are performed under noise, reverberation and spontaneous conversational turn-taking [9].

These datasets are foundational to ASR research because they retain the properties that are essential for downstream evaluation and application, like realistic microphone recording conditions, speaker interruptions, incomplete sentences and natural overlap. They together represent the trade-off between ecological validity and controlability. The real datasets preserve natural overlap patterns, acoustic effects and turn-taking, but they do not allow independent control of degradation factors. Inversely, synthetic datasets allow for precise control of factors but often lose the realism of natural conversation. This dissertation is designed to contribute to the middle ground, balancing realism and control.

Therefore, synthetic mixture generation has become a common technique in robust speech research. Earlier speech-separation datasets were often built by randomly mixing utterances under simplified conditions, but later datasets have progressively introduced realism by adding environmental noise, reverberation, corpus diversity and lexical continuity. *WHAM!* added real ambient noise and argued that clean, fully overlapped benchmarks were too narrow for realistic evaluation [10]. *WHAMR!* enhanced this by adding reverberation, modelling the spectral and temporal smearing that arises in indoor environments [11]. *LibriCSS* then provided a more generalised alternative to wsj0-2mix-based datasets like *WHAM!* with mixture derived from

LibriSpeech [3]. Finally, *LibriMix* extended this by mixing in VCTK speech, increasing the total length and thereby making it more suitable for training purposes [12]. The field has provided a rich set of resources for ASR research, but there is yet to be a framework that allows customised manipulation of overlap, noise, duration and speaker count in a reuseable way for generating synthetic mixtures.

2.1.2 Overlap Modelling and Evaluation Metrics for Multi-Speaker ASR

In this literature, the overlap ratio is a useful way to measure and control mixture difficulty. In LibriCSS, the overlap ratio is defined as the total duration of the overlapped region divided by the total speech duration [3]. This is methodologically important because natural conversation is not uniformly overlapped. It is often varied between different dialogue contexts [13]. A benchmark that reduces these contexts into a single average condition hides what a model is actually robust to and what it is not. The evolution from a simple fully overlapped mixture to realistically partially overlapped speech signified an important progress in realism, showing the increasing emphasis on treating overlap as an experimental variable rather than a marginal condition [14].

In ASR evaluation, the choice of metric can influence the weights of different aspects considered. In single-speaker ASR, word error rate (WER) is the standard metric, but in multi-speaker and overlapping settings, its calculation and interpretation are problematic. Galibert [15] shows that classical implementations of WER and diarisation error rate (DER) that were designed for single-speaker speech need to be generalised and adapted to multi-speaker conditions to remain meaningful. Several recent works have formalised the metric for multi-speaker transcription in different aspects. von Neumann et al. [16] shows that variants like cpWER or ORC-WER can be understood within a wider framework of long-form multi-speaker WER, while von Neumann et al. [17] provides a toolkit for computing metrics like WER, cpWER, MIMO-WER and OCR-WER. von Neumann et al. [16] also discusses the separation between speaker-attributed metrics and speaker-agnostic metrics by comparing asclite, OCR-WER, cpWER and MIMO-WER. The conclusion is that lexical recognition, temporal information and speaker attribution are all important factors in multi-speaker ASR evaluation, and that no single metric captures all of these aspects equally well. As a result, the choice of metrics must therefore be aligned with the application and the specific aspect of robustness being evaluated while methods of comparing mismatched reference and hypothesis structures remain underexplored.

2.1.3 ASR Architectures and Benchmark Systems for Robust Speech Transcription

The model architecture operating on top of these datasets and metrics has evolved substantially. One family of models is the self-supervised encoders like wav2vec-2.0, which show that pretraining on very large unlabeled examples can learn representations that transfer effectively to downstream ASR by fine-tuning on smaller labelled datasets [18]. Another family is the large encoder-decoder models like Whisper, which uses weakly supervised pretraining and focuses on zero-shot transfer to improve robustness [19]. The third family is the conformer and transducer-based architectures like Parakeet that augmented the transformer with convolutional modules to capture both local patterns and global interactions and widely used in streaming ASR [20]. In modern multi-speaker ASR research, these models are often embedded inside complex pipelines that include front-end separation, speaker diarisation and transcription modules [21, 22]. In conclusion, the field already has mature corpora, mixture-generation, evaluation metrics and ASR models. However, the systematic framework for isolating the effects of overlap, noise and other factors under controlled environments remains less mature.

2.2 Establishing a Gap in the Literature

Despite the progress in multi-speaker ASR research, there are still important methodological gaps for this dissertation. Current real meeting corpora are ecologically valid, but they do not function as reusable testbeds in which overlap ratio, signal-to-noise ratio (SNR), clip duration, and speaker count can all be varied independently and systematically [1, 9]. On the other hand, synthetic datasets each solve only part of the problem. For example, WHAM! added noise, WHAMR! added reverberation, LibriMix provided more data and more realistic mixtures [10, 11, 12]. However, none of these resources was designed primarily as a configurable framework for causal analysis of ASR robustness. Hence, they provided many excellent benchmarks, but less support for answering narrow experimental questions about the standalone and interactive effects of degradation factors on ASR performance.

A second gap concerns the evaluation under structural mismatch between references and hypotheses. Recent metric work has clarified how to judge speaker-attributed hypotheses, speaker-agnostic hypotheses and multi-output systems [16, 17, 23]. However, in practice, many strong ASR systems still output a single sequential hypothesis even when the reference is multi-speaker, while overlap benchmark often encode reference speech as concurrent stream [18, 20]. This creates an awkward in-between case. While cpWER and tcpWER are designed for multi-stream references, they assume a speaker-structured hypothesis. ORC-WER is designed to ignore speaker labels and isolate lexical correctness. However, what is not provided is an estimation algorithm tailored to the specific case of comparing a double-stream reference to a single-stream hypothesis.

This dissertation will therefore address two gaps. The first is the absence of a reusable, reconfigurable synthetic mixture generation framework for overlapping speech that can manipulate overlap, noise, duration and speaker count as experimental variables. The second is the absence of an estimation algorithm for the particular evaluation scenario in which a potentially overlapping double-stream reference must be compared with a single-stream hypothesis transcript. The proposed solution is motivated by the second gap. It is to make the evaluation of mismatched reference and hypothesis structure more informative and fair. In conclusion, these problems define the niche of this dissertation: controlled robustness benchmarking for overlapping speech and an evaluation method for mismatched reference and hypothesis structures.

2.3 Recent Advances

2.3.1 Realism and Generalisation in Overlap Modelling

Recent work has attempted to reduce some of these limitations. Some research has focused on improving the realism of synthetic overlap. Yang et al. [14] criticise the standard practice of randomly mixing utterances and propose a more realistic simulation by learning overlap patterns from real conversation, and representing overlap patterns as a sequence of discrete tokens. Their results show that the ASR model trained with a realistic overlap simulation consistently outperforms models trained with random overlap. In the benchmarking direction, Cornell et al. [24] spurs the generalisation across speaker counts, formal vs informal settings, meeting duration, acoustic scenarios and recording configurations. Although they have made valuable progress from optimisation under specific conditions to broader generalisation, improving realism, they have yet to provide systematic frameworks to control overlap, SNR, duration, and speaker count in a clean design.

2.3.2 Integrated Pipelines and End-to-End ASR for Overlapping Speech

A second direction of work has concentrated on integrated meeting-recognition pipelines. [21] shows that an end-to-end modular system that integrates independently trained speech separation,

diarisation and transcription modules can achieve competitive performance on LibriCSS. Chen et al. [25] demonstrate that the combination of continuous speech separation (CSS) and conformer-based ASR can achieve state-of-the-art performance in both single-channel and multi-channel settings on LibriCSS. von Neumann et al. [17] use a pipeline that combines the CSS system with TF-GridNet separation and a speaker-agnostic ASR and achieves outstanding performance on ORC-WER. They also propose a syntactically informed diarisation for turn detection, which results in exceptional performance on cpWER. The strength of these modules is that they treat meeting transcription as an all-in-one problem rather than a pure ASR problem. This, conversely, brings the drawback of ambiguity on the source of improvements, which makes it harder to locate the specific factors that contribute to the improvements and also makes it harder to locate the point of failure.

A third line of work has turned to an end-to-end ASR framework for speaker-attributed modelling. Kanda et al. [26] present a transformer-based end-to-end model that performs speaker counting, speech recognition and speaker identification, reducing the dependence on separated separation and diarisation modules. Chang et al. [27] shows that such a model is susceptible to the mismatch between training and testing conditions and introduces a hypothesis stitcher that fuses multiple hypotheses into a single output to help improve accuracy. Kanda et al. [28] push this further by showing speaker-attributed transcription can be used to derive diarisation output, achieving a comparable diarisation error rate to existing diarisation methods and better if the number of speakers is not known. Conclusively, these studies are important because they treat lexical content and speaker identity as mutually informative independent tasks. On the flip side, they also reveal that since these systems are more specialised, the training assumptions are stronger, and the resulting outputs are less comparable to single-stream ASR baselines.

More recent papers have tried to reduce the gap between theoretical models and practical applications. Cui et al. [29] propose a pipeline for real-life applications that involves voice activity detection (VAD), speaker diarisation (SD), and speaker-attributed ASR (SA-ASR). They show that using VAD results to finetune the SA-ASR model significantly reduces the speaker error rate (SER) and demonstrate that extracting speaker embedding from SD can further improve the SER. Raj et al. [30] push the SURT framework from speaker-agnostic to speaker-attributed ASR by adding a speaker branch and prefixing each chunk with speaker information. Zheng et al. [31] introduces the DNCASR model for joint neural speaker clustering and ASR, effectively addresses overlapping speech and achieves strong cpWER performance on AMI. These studies indicate that the field is moving towards more application-oriented systems of long-form, speaker-consistent and meeting-level transcription. Nonetheless, they still rely on a small set of benchmarks. They therefore match the need for studies that evaluate ASR systems under different controlled conditions.

2.3.3 Unified Metrics for Multi-Speaker ASR Evaluation

Recent literature on ASR metrics provides evolving insights into multi-speaker evaluation. von Neumann et al. [17] contribute to standardising the computation of WER, cpWER, ORC-WER and MIMO-WER as a toolkit. von Neumann et al. [23] contributes to the conceptual framework for clarifying which metrics are suitable for which output structure and which aspect of performance. They also propose efficient approximation algorithms for expensive metrics like ORC-WER and DI-cpWER. These works discourage the default use of a single metric for outputs that embody different assumptions about speaker structure and timing. However, even this more mature scoring literature still treats the metrics primarily as ways of assessing matching systems. It does not directly solve the narrower experimental problem of how to compare double-stream references with single-stream hypotheses. That is, therefore, the space in which the present study makes its methodological contribution.

2.3.4 Summary

Overall, parallel research has advanced in three main directions. It has made the synthetic overlap more realistic rather than purely random. It has strengthened meeting transcription through integrated modular pipelines that combine separation, diarisation, and recognition. It has progressed ASR evaluation metrics towards a more comprehensive approach. However, it still leaves a gap in not having a lightweight, reusable framework for systematically varying degradation factors and the metrics that handle stream-mismatched evaluation. The present dissertation is designed to contribute to that missing middle ground.

Chapter 3

Architectural Design

3.1 Design Objective and Principles

This chapter describes the software architecture for synthetic mixture generation, automatic speech recognition (ASR) inference under controlled identical conditions, evaluation using both established and proposed transcription metrics, and statistical analysis. The following principles guide the design:

- Experimental control: overlap ratio, signal-to-noise ratio, clip duration, speaker count and noise type must be varied independently and systematically.
- Controlled comparison: ASR models must be evaluated under the same set of synthetic mixtures with model-specific decoding.
- Traceability: every result must be traceable to the specific configuration, including source files, offsets, transcripts and inference outputs.
- **Modularity**: all components, including mixture generation, ASR inference, metric computation and statistical analysis, should be self-contained and separable so that new models or metrics can be added without rewriting the system.
- **Reproducibility**: random seeds, libraries and model versions must be recorded so that other researchers can reproduce the experiment.

3.2 System Overview

Figure 3.1 depicts the main pipeline of the system. The system starts from clean speaker audio folders (containing audio from a single speaker, with the speaker ID as the folder name and the corresponding transcripts) and noise audio folders (.wav or .flac, with the noise type as the folder name). A configuration file specifies experimental factors for mixture generation, including the number of mixtures, SNR levels (SNR), overlap ratios, duration, and speaker count (number of unique speakers). The *Mixture Generator* module creates synthetic mixtures based on these configurations and outputs both the audio files and a manifest (containing metadata and transcripts). The mixed audio is fed into the target ASR models for inference. The result hypotheses are scored against the reference transcripts using evaluation metrics. The resulting hypotheses are then evaluated against the reference transcripts using specified evaluation metrics. The final results are analysed and compiled into a final report.

3.2.1 System Boundary and Design Rationale

Table 3.1 summarises the main components of the system and their responsibility. A modular architecture is adopted to allow for flexibility and extensibility, such that each component can

Table 3.1: Summary of the main components in the experimental framework.

Component	Input	Output	Responsibility
Synthetic mixture generation	Overlap ratio, SNR, noise type, speaker count	Synthetic audio, transcripts, and manifest	Generate synthetic audio mixtures that satisfy the specified acoustic and structural conditions.
DSS-WER beam approximation	Double-stream reference and single-stream hypothesis	DSS-WER approximation	Approximate a word error rate that permits interleaving between the two reference streams.
Benchmarking	None	Time statistics for four models	Measure the inference-time cost of each ASR model under a common benchmarking procedure.
Statistical notebook module	Transcriptions and WER values for each model and clip	Statistical plots and tables	Transform raw transcription outputs into statistical summaries, comparisons, and interpretable empirical claims.

function independently. The synthetic mixture generation is separated from the ASR inference to ensure that synthetic mixture generation can also be used for other purposes such as training ASR models, and to allow for the possibility of adding new ASR models without modifying the generation code. The metric computation and statistical analysis are separated to allow for the addition of new metrics or new statistical analyses without modifying the inferencing code. This also keep the raw data so that new analyses can be performed without rerunning the entire system.

3.3 Artifact Structure

3.3.1 Synthetic Audio Manifest Structure

Every clip is recorded in a manifest as a CSV file with metadata and transcripts. The transcript is a sequence of tuples of the form (speaker id, text, start time, end time) for each utterance. Table 3.2 summarise the key fields.

3.3.2 Transcription Output Schema

The output transcription JSON file has the following format:

- **key**: clip ID
- **transcript**: a dictionary of the form `{model name: hypothesis}` for each ASR model, where each hypothesis is a list of tuples of the form (speaker id, text) for each utterance.

3.3.3 Metric Output Schema

The evaluation results are stored in a JSON file with three top-level fields: `summary`, `results`, and `timestamp`. The `summary` field contains aggregate information over the full evaluation run, including the total number of clips processed, the numbers of successful and failed evaluations, and descriptive statistics for the reported metrics. In particular, it stores counts and summary statistics for standard word error rate (WER), DSS-WER, and ORC-WER.

Field	Purpose	Example
`clip_id`	Unique key joining audio, predictions, metrics, and statistics.	`mix_0000104_0.00_2_0_T`
`audio_path`	Relative path to the generated waveform.	`output/audio/mix_000031.wav`
`transcript`	Ordered list of utterances with speaker ID, text, and timestamps.	`[(spk,text,t0,t1),...]`
`source_files`	Provenance of the clean speech segments used.	list of input `.flac` or `.wav` paths
`noise_files`	Provenance of the noise segment(s) used.	list of noise `.flac` or `.wav` paths
`speakers_count`	Number of distinct speakers in the mixture.	`2`
`overlap_ratio_target`	Requested overlap ratio from configuration.	`0.20`
`overlap_ratio_actual`	Realised overlap ratio after scheduling.	`0.20`
`snr_db`	Requested SNR, or `clean` for no added noise.	`7.4`
`noise_type`	Noise category for condition grouping.	`P` (public)

Table 3.2: Recommended manifest schema. The key design principle is that every downstream analysis should be derivable from `clip_id` plus stored metadata.

The `results` field contains a list of per-clip evaluation records. Each record corresponds to one audio clip and takes one of two forms. For an unsuccessful evaluation, the record instead stores the clip identifier, model name, evaluation status, and an error message describing the failure.

Finally, the `timestamp` field records when the output file was generated. This structure separates corpus-level summary statistics from clip-level measurements, while also preserving failure information for later inspection and debugging.

3.4 Synthetic Mixture Generation

The synthetic mixture is the core module of the system. It determines how overlap and noise are controlled. It comprises three steps: (1) **Speaker and clip selection**: randomly select the required number of speakers and clips from the clean speech dataset, (2) **Offset scheduling**: (1) **Offset computation**: calculate the time offsets for each utterance such that the overall overlap ratio approaches the target. (2) **Residual adjustment**: adjust the time offsets calculated to ensure the overall overlap ratio exactly matches the target. (3) **Noise addition**: add noise to the mixture at the specified SNR level. The output of this process will be a set of audio with different speaker counts and overlap ratios as base clips, and each base clip will have variants with different SNR levels. The metadata and transcripts of the mixtures are stored in a manifest file.

3.4.1 Speaker and Clip Selection

This *clip-selection* process takes a target speaker count c and a duration d (minimum duration of final mixed audio), and returns: (1) a sequence of clips satisfying the target speaker count and duration constraints, (2) the corresponding sequence of transcripts.

Figure 3.2 shows the process that is designed to create a sequence of clips with specified speaker count and duration. A set of c speakers is selected at uniform random without replacement. The process iterates until the total duration of the selected clips exceeds the target duration adjusted by the overlap ratio (to account for the overlapping region that is double-counted here). The adjusted total clip duration T_{src} is calculated by:

$$T_{\text{src}} = T_{\text{active}} + T_{\text{overlap}}. \tag{3.1}$$

If the overlap ratio is defined as

$$\text{OVR} = \frac{T_{\text{overlap}}}{T_{\text{active}}}, \tag{3.2}$$

Then the required total source duration is

$$T_{\text{src}} = d(1 + \text{OVR}) \tag{3.3}$$

This formulation is preferable to the naive duration threshold because it accounts for the double-counted time by the overlapping region. Without this, a highly overlapped mixture would have a much shorter duration than the intended target.

A balanced selection is used during the clip selection process. The next speaker is chosen with a probability inversely proportional to the number of clips contributed by each speaker, to balance the number of clips per speaker (with probability 0 for speakers from the previous iteration). The random selection of clips is uniform random with replacement. This process mimics an actual conversation, with speakers taking turns.

3.4.2 Offset Scheduling

This scheduler given a sequence of clips with durations $\ell_1, \ell_2, \ldots, \ell_n$ and overlap ratio ($\text{OVR} \in [0, 1]$), constructs clip starting sample index such that the overlap ratio is exactly as specified, using the given overlap std ratio σ_{OVR}, and clip spacing mean and std (for non-overlapping case).

In the case where the overlap ratio is zero, the start time of each clip is determined by allocating a spacing that is a random sample from a normal distribution $\mathcal{N}(\mu, \text{std})$ where μ and std are the mean and std of the clip spacing specified. In the case where the overlap ratio is greater than zero, the target absolute overlap o^* is

$$o^* = \frac{\text{OVR}}{1 + \text{OVR}} \cdot \sum_{i=1}^{n} \ell_i. \tag{3.4}$$

The total is distributed across the adjacent clips. Let o_i be the overlap between clip i and clip $i + 1$, then the scheduler enforces

$$0 \leq o_i \leq \min(\ell_i, \ell_{i+1}), \qquad \sum_{i=1}^{n-1} o_i = o^*. \tag{3.5}$$

For n clips, the average overlap is

$$\bar{o} = \frac{o^*}{n - 1} \tag{3.6}$$

then each overlap o_i is sampled as

$$o_i = \mathcal{N}(\bar{o}, (\bar{o} \cdot \sigma_{\text{OVR}})^2) \tag{3.7}$$

After that, a residual correction is applied so that the sum of sampled overlaps equals the target overlap exactly. The difference diff $= o^* - \sum$ sampled overlaps. If the difference is greater than zero, we randomly select overlaps that are smaller than the preceding and succeeding clip durations, and add one sample to each selected overlap until the difference is zero. If the difference is negative, we randomly select overlaps greater than zero and subtract one sample until the difference is zero. Finally, the start sample index of each clip is determined by adding the spacing (overlap will be treated as negative spacing).

The start times s_i are then computed recursively as

$$s_1 = 0, \qquad s_i = s_{i-1} + \ell_{i-1} - o_{i-1} \quad \text{for } i = 2, 3, \ldots, n. \tag{3.8}$$

An important architectural design choice here is treating the overlap as a controlled variable. This makes the generator suitable for experimental work rather than just producing plausible mixtures.

3.4.3 Noise Addition

After the clean mixture is generated, environmental noise is added.

The noise addition process takes input of the mixed audio, set of noise audio, noise type and target SNR level ($\mathrm{SNR}_{dB} \in \mathbb{R}$ or clean, measured over active speech time). The noise type can be domestic (D), public (P), transportation (T), nature (N), office (O), street (S), or all (A) (see Section 4.4.1 for more details).

Then, noise audio samples are uniformly sampled from the specified noise type(s) and concatenated until the total length is reached, with excess parts truncated. Let $A_{\text{active speech}}$ be the root mean square (RMS) amplitude of the active speech region, and let A_{noise} be the RMS amplitude of the noise segment before scaling. For a target SNR level of SNR_{dB} defined as

$$\mathrm{SNR}_{dB} = 20 \log_{10} \left(\frac{\mathrm{A}_{\text{active speech}}}{\mathrm{A}_{\text{noise}}} \right), \tag{3.9}$$

The noise is scaled by a factor of

$$\alpha = 10^{-\frac{\mathrm{SNR}_{dB}}{20}} \cdot \frac{A_{\text{active speech}}}{A_{\text{noise}}} \tag{3.10}$$

The active speech region is determined by the RMS of all clips in the mixture.

The final mixture is the sum of the clean mixture and the scaled noise.

$$x_{\text{final}}(t) = x_{\text{speech}}(t) + \alpha x_{\text{noise}}(t) \tag{3.11}$$

The SNR is measured relative to the active speech region to ensure the noise level is accurately proportional to the speech content, preventing long silence regions from distorting the effective noise level.

This algorithm is designed to support controlled experiments on overlap ratio, SNR level, speaker count, duration and noise type by varying each factor independently. It also supports noise analysis on the same clip with different noise types and SNR levels.

3.5 ASR Inference and Experiment Orchestration

The inference layer consumes the manifest and produces standardised hypotheses for each ASR model. Each ASR backend is wrapped by a common adapter interface, enabling uniform orchestration across models. The wrapper is responsible for:

1. Periodic GPU cache clearing and optional garbage collection to manage memory during large-scale inference.

2. Skipping already transcribed clips by checking the existing JSON file
3. Running inference over a batch of clips
4. Serialising the output into the standardised JSON format
5. Recording runtime and failure information

The adapter is important because the ASR models are heterogeneous. Some return segmented outputs, while some return unsegmented outputs. Some inputs are audio waveforms, while some take file paths. This architecture, therefore, standardises the output format and provides a unified interface. To support resumability, the wrapper checks whether a clip has already been transcribed before running inference. Exceptions are logged per batch, so a single clip failure does not break the entire process. This is practical as when large experiments are executed over days, robustness and resumability are important to prevent loss of progress.

3.5.1 Fairness Controls

The following controls ensure that measured differences are attributed to the ASR models rather than confounding factors:

- **fixed sample rate**: 16kHz for all models, resampling is applied if necessary
- **identical audio**: the same set of audio files is used across all models
- **identical reference**: the same set of reference transcriptions is used across all models
- **text normalisation**: the same text normalisation is applied to both reference and hypothesis before evaluation, including lowercasing and punctuation removal.

3.6 Evaluation Architecture

The evaluation system is a separate architectural block. This is where the dissertation's scientific contribution lies.

3.6.1 Established Metric Modules

The first module computes established metrics, such as WER for single-stream referencing and ORC-WER for multi-stream referencing.

WER

Given a reference transcript and a hypothesis transcript, WER is computed as the Levenshtein distance between the reference and hypothesis:

$$\text{WER} = \frac{\text{lev}(r, h)}{|r|} = \frac{I + D + S}{|r|} \tag{3.12}$$

where I, D and S are the number of insertions, deletions and substitutions required to transform the hypothesis into the reference, and $|r|$ is the number of words in the reference [7].

ORC-WER

The ORC-WER is speaker-agnostic. It computes the minimum WER over all possible segments-to-utterance assignments. It is defined as

$$\mathrm{lev}^{(\mathrm{ORC})} = \min_{\rho_1,\ldots,\rho_J} \sum_{j=1}^{J} \mathrm{lev}\left(\bigoplus_{u\in\rho_j} \mathcal{R}_u,\ \bigoplus_{u'=1}^{U_{\mathcal{H},j}} \mathcal{H}_{j,u'} \right) \tag{3.13}$$

where rho_j contains index pairs of speaker i and utterance u, meaning reference utterance u from speaker i is assigned j-th hypothesis segment. $\bigoplus$ is the concatenation operator.

3.6.2 Proposed Metric Module: Double-Stream Serialised WER (DSS-WER)

The DSS-WER is a proposed metric that extends WER to solve the specific case of double-stream reference transcriptions with single-stream hypotheses. Given two reference streams A and B representing two speakers, and a single-stream hypothesis H, the DSS-WER computes the minimum WER between the hypothesis and any interleaving merging of the two reference streams. Formally, let A and B be a partial order of words, and let R be the set of all possible interleaving mergings of A and B that preserve the partial order. Then the DSS-WER is defined as

$$\text{DSS-WER}(A, B, H) = \min_{r\in R} \mathrm{WER}(r, H). \tag{3.14}$$

Example of DSS-WER vs. WER

Figure 3.4 shows an example where the ordinary concatenated WER and the DSS-WER differ. DSS-WER is able to give zero error for a valid interleaving of the two reference streams, whereas WER penalises the different serialisation order. This illustrates the important distinction between the two metrics and the motivation for the proposed DSS-WER.

However, since the number of possible interleavings grows combinatorially with the length of the reference, the time complexity of computing DSS-WER exactly is $O\left(\binom{|A|+|B|}{|A|} \cdot (|A| + |B|) \cdot |H|\right)$, which is computationally infeasible for long transcripts. Therefore, an efficient approximation algorithm is necessary for practical use. Therefore, we hereby propose an efficient approximation algorithm for DSS-WER.

3.6.3 Beam-Search Approximation for DSS-WER

To evaluate a single hypothesis transcript against two reference speaker streams while allowing interleaving, I implement a beam-search approximation to the minimum levenshtein distance over all valid merges of the two reference chains. Figure 3.5 is the overall flowchart. The full pseudocode is in Appendix A.

Let

$$A = (a_1, a_2, \ldots, a_m), \qquad B = (b_1, b_2, \ldots, b_n), \qquad H = (h_1, h_2, \ldots, h_T) \tag{3.15}$$

denotes the two reference chains and the hypothesis chain.

Normalisation

Each word in the reference and hypothesis is normalised by lowercasing and removing punctuation. This is to ensure that the metric is not affected by mismatches caused by case or punctuation differences rather than recognition errors.

State Definition

A search state is defined as

$$s = (i, j, t), \tag{3.16}$$

where i is the number of words consume from chain A, j is the number of words consumed from chain B, and t is the number of words consumed from the hypothesis H. The initial state is $(0, 0, 0)$ and the terminal states are (m, n, T).

Transition and Cost

For non-terminal states $s = (i, j, t)$, there are up to 5 transitions:

1. **Match / Substitution from** A: if $i < m$ and $t < T$, align a_{i+1} with h_{t+1}, gives

$$(i, j, t) \to (i + 1, j, t + 1), \tag{3.17}$$

   The cost is 0 if $a_{i+1} = h_{t+1}$ and 1 otherwise.

2. **Match / Substitution from** B: if $j < n$ and $t < T$, align b_{j+1} with h_{t+1}, gives

$$(i, j, t) \to (i, j + 1, t + 1), \tag{3.18}$$

   The cost is 0 if $b_{j+1} = h_{t+1}$ and 1 otherwise.

3. **Insertion**: if $t < T$, insert h_{t+1} into the merged sequence, gives

$$(i, j, t) \to (i, j, t + 1), \tag{3.19}$$

   The cost is 1.

4. **Deletion from** A: if $i < m$, delete a_{i+1}, gives

$$(i, j, t) \to (i + 1, j, t), \tag{3.20}$$

   The cost is 1.

5. **Deletion from** B: if $j < n$, delete b_{j+1}, gives

$$(i, j, t) \to (i, j + 1, t), \tag{3.21}$$

   The cost is 1.

These transition preserves the partial order of the two reference chain while allowing any interleaving.

Scoring Function

Each state s maintains the best exact cost $g(s)$ while candidate states are ranked using

$$f(s) = g(s) + \lambda h(s), \tag{3.22}$$

where $\lambda \geq 0$ is a tunable heuristic weight, and $h(s)$ is a lower bound estimate of the remaining cost.

Heuristic Function

The heuristic is the maximum of the following two estimates:

- **minimum remaining edit distance**:

$$h_{\text{length}}(s) = |((m-i) + (n-j) - (T-t))| \tag{3.23}$$

  is the minimum number of insertions or deletions required to consume all remaining words from the references and hypothesis, assuming perfect matches.

- **local bag of words lower bound**:

$$h^{\ell}_{\text{bow}}(s) = \left\lceil \frac{1}{2} \cdot \sum_{w} |\text{count}(w, A_{i+1:i+\ell}) + \text{count}(w, B_{j+1:j+\ell}) - \text{count}(w, H_{t+1:t+\ell})| \right\rceil \tag{3.24}$$

  where $\text{count}(w, S)$ is the number of occurrences of word w in sequence S, and ℓ is the lookahead window size (truncated to the end of the sequence). This is a lower bound because it assumes that the best case where all matching words in the lookahead window can be aligned, and the remaining unmatched words can be perfectly aligned.

The heuristic is then

$$h(s) = \max(h_{\text{length}}(s), h^{\ell}_{\text{bow}}(s)), \tag{3.25}$$

preserving admissibility and take the stronger approximation.

Beam Pruning

At each iteration, all successor states are ranked by $f(s)$ with ties breaker of $g(s)$. The algorithm only keeps a bounded number of candidates. The implementation has two main pruning strategies:

- **adaptive beam size**: the beam size is dynamically adjusted based on the distribution of $f(s)$ scores. Given an ambiguity threshold τ, base beam size B and maximum beam size $B_{\max}$, if number of candidates with $f(s) \leq f_{\text{best}} + \tau$ is greater than $\frac{B}{2}$, beam size is set to $B_{\max}$, and B otherwise. This allows search to be more expansive when best path is more ambiguous, and more aggressively when best path is more clear.

- **stratified beam pruning**: candidates are stratified into different buckets based imbalances in consumption in chain A and chain B (i.e. $i - j$), and a balanced number of candidates are kept from each group. The buckets are defined by $\lfloor (i-j)/\max(1, \text{bucket width}) \rfloor$ (where bucket width is a tunable parameter), then the buckets are ranked by the best $f(s)$ score of each bucket, and candidates are selected in a round-robin manner from the ranked buckets until the beam size is reached. This is to prevent the beam from dominated by similar alignment and increase the diversity of the search.

Bidirectional Search

The search is run in both forward and reverse direction, and the best result from both directions are returned. The reverse direction search can discover better alignment that might be pruned in the forward direction.

Final Output

For best path distance d^* found from both directions, the DSS-WER is computed as

$$\frac{d^*}{m+n} \tag{3.26}$$

In addition, the best path alignment is also returned as a list of tuples of the form (operation, chain used, reference word, hypothesis word). This allows downstream error analysis and hyperparameter tunings.

Complexity

Let

- ℓ be the lookahead window size for the heuristic,
- B be the maximum beam size,
- E be the maximum number of expansions (which is the upper bound of the expansions during the search)

The time complexity is

$$O(E(\ell + B \log B)), \tag{3.27}$$

where there is E expansions, each expansion generates up to 5 sucessors, and each successor requires $O(\ell)$ for heuristic computation and $O(B \log B)$ for sorting during beam pruning. The space complexity is $O(E)$ for storing all expanded states.

3.6.4 Metric Output Schema

All metric results are stored at clip level in a JSON file with the following fields: `clip_id`, `model`, `wer`, `orcwer`, `dsswer`.

3.7 Benchmarking

Inference-time benchmarking was used to compare the operational cost of the ASR systems under a common evaluation protocol. A fixed subset of 100 clips was sampled from the experiment manifest using a fixed random seed, and each model was run on the same sampled set.

The runtime metric recorded was measured using `time.perf_counter()` to capture the total wall-clock time taken for inference on the sampled set. This captures the end-to-end time instead of just the decoding time, as it reflects the actual user experience and operational cost.

The time taken per clip was computed by dividing the total runtime by the number of clips in the sampled set, while the Inverse Real-Time Factor (RTFx) was computed as:

$$\text{RTFx} = \frac{\text{total audio duration}}{\text{Processing time}} \tag{3.28}$$

Execution was performed on CUDA. The implementation used batch size 100 `parakeet` as it allows batch inferencing, while the other systems were evaluated sequentially on the sampled clips. GPU cache clearing and garbage collection were applied between runs, but GPU memory usage was not logged as an output statistic. Likewise, failed runs were not recorded through a separate structured logging system.

3.8 Real-Data Segmentation and Evaluation Pipeline

A separate pipeline was implemented to evaluate ASR performance on real conversational audio. Speaker-labelled utterances were first loaded from a JSON transcript, converted into timestamped intervals, and sorted chronologically. Fixed-length clips were then constructed from valid utterance boundaries so that no reference utterance was truncated across clip limits.

For each extracted clip, the corresponding waveform segment and reference transcript were written to disk as reusable evaluation artefacts. These clips were then transcribed by the selected ASR models under a common batching procedure, with intermediate outputs cached to JSON in order to support recovery from interrupted runs.

Evaluation was performed clip by clip against the exported references. Where a model returned segmented output, ORC-WER was used; otherwise, standard WER was applied to the flattened hypothesis. The pipeline therefore provided a unified framework for comparing heterogeneous ASR systems on real data while preserving both per-clip results and aggregate summary statistics.

3.9 Statistical Analysis Module

All results from transcription and metric evaluation are stored in a clip-level table indexed by clip ID and model. This table is used in the statistical analysis module.

This module converts raw clip-level scores into statistical claims about the generator and ASR models. The main functionalities include:

- **Descriptive statistics**: summarise the distribution of scores under each model and condition
- **Paired comparisons**: perform paired statistical tests between models evaluated on the same set of clips
- **DSS-WER and WER comparison**: perform paired comparisons between DSS-WER and WER to evaluate the proposed metric

3.10 Design Trade-offs and Limitations

This section discusses the design trade-offs and limitations of the architectural choices.

- **Controlled generation vs natural realism**: the generator is designed to support precise control over factors. This is ideal for scientific experiments but may results in less natural mixtures compared to using actual conversation.
- **Metric accuracy vs simplicity**: the DSS-WER approximation algorithm improves accuracy in capturing transcription errors in double-reference-single-hypothesis cases. It better reflects the transcription quality, but it is more complex and substantially more computationally expensive than WER.

3.11 Reproducibility and Experimental Setup

System Environment

Operating System Ubuntu 22.04

Python Version 3.14.3

GPU NVIDIA GeForce RTX 4070 Laptop GPU with 8GB VRAM

Random Seed 1234

Software dependencies and configuration details are documented in Appendix B

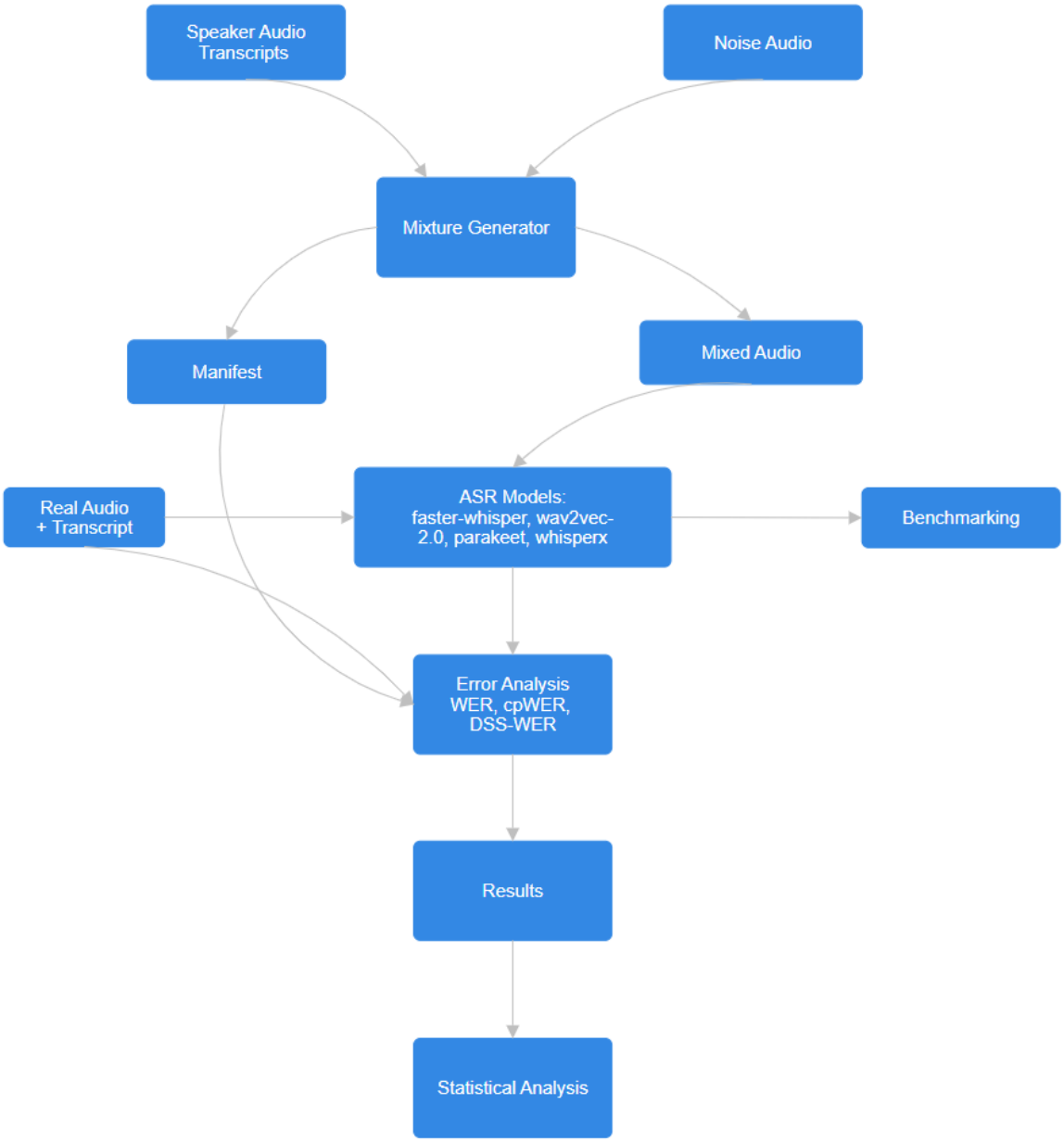


Figure 3.1: High-level data flow through the experimental system. Clean speech and noise recordings were transformed into synthetic mixtures, transcribed by ASR models, and passed to evaluation, benchmarking, and statistical analysis. (Generated with SmartDraw [32].)

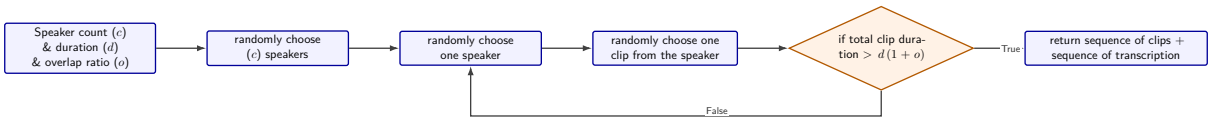


Figure 3.2: Speaker and Clip Selection Process. The stopping criteria are based on the required source duration, which is adjusted by the overlap ratio to account for occupied time by the overlapping region.

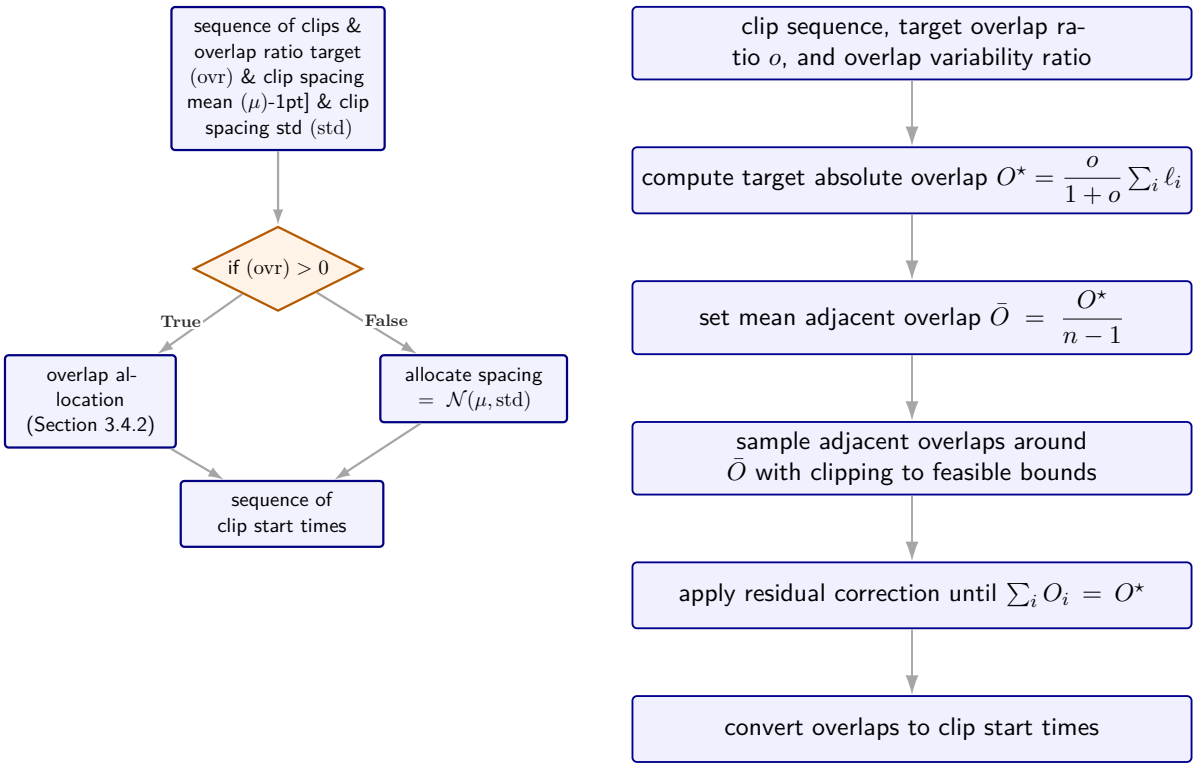
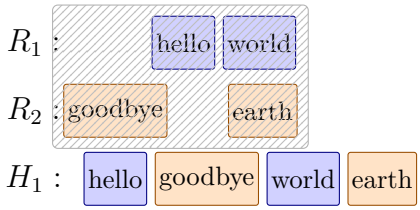
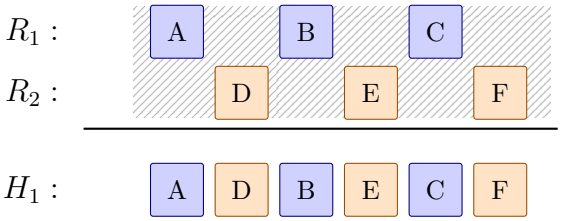


Figure 3.3: Offset Scheduling Process. The left diagram shows the top-level decision flow between non-overlapping and overlapping cases. The diagram on the right shows the overlap allocation process when the overlap ratio is greater than zero.



Metric	Error
WER	1/4
DSS-WER	0/4

(a) A toy lexical example in which the hypothesis forms a valid interleaving of the two reference streams.



Metric	Score
WER	2/6
DSS-WER	0/6

Example: Ordinary WER can over-penalise a hypothesis that is a valid interleaving of multiple reference streams. Here, H_1 = `A D B E C F` preserves the alternating order implied by R_1 and R_2, so DSS-WER assigns zero error. In contrast, ordinary WER compares the hypothesis against a single serialised reference and yields a non-zero score.

(b) An alternating interleaving example where ordinary WER and DSS-WER produce different scores.

Figure 3.4: Examples illustrating the difference between ordinary WER and an interleaving-aware metric. In both cases, the hypothesis is a valid interleaving of the reference streams, so DSS-WER assigns zero error, whereas ordinary WER penalises the mismatch against a single serialised reference.

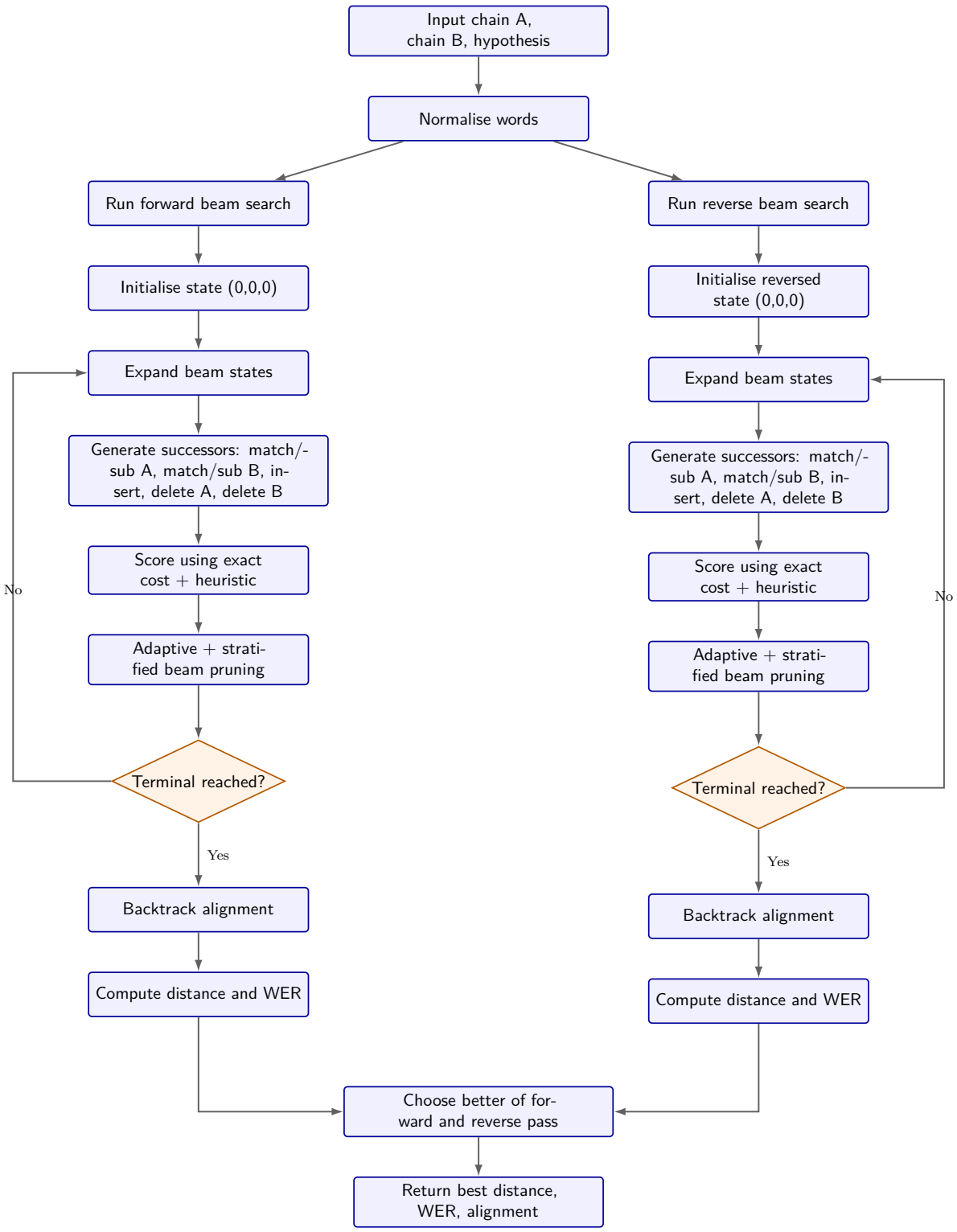


Figure 3.5: High-level flow of the beam-search approximation algorithm for DSS-WER.

Chapter 4

Experimental Design

4.1 Objective and Scope

This chapter specified the empirical design used to evalute the robustness of multiple automatic speech recognition (ASR) transcription models under overlapping speech and addition noise, and to assess the utility and accuracy of the newly proposed Double-Stream Serialisation WER (DSS-WER) when two streams of reference are compared against a single hypothesis. The evaluation is grounded in the structures and pipelines proposed in Chapter 3 and is divided into three tracks.

The first track is the controlled synthetic benchmarking of ASR performance in which overlap ratio, SNR and noise type are manipulated independently. This provides a setting where degradation factors can be isolated and their effects on ASR performance can be causally attributed, while the ASR models can be evaluated under identical conditions for fair comparison. The second track is a real-world data transferability evaluation on CHiME-6, which assesses whether the trends observed in the controlled synthetic mixtures are visible in natural conversational audio with realistic overlap and noise conditions. The third track is a hypothesis testing of the proposed DSS-WER metric, which evaluates whether it provides additional insights and accuracy in quantifying ASR performance under overlapping conditions compared to standard WER.

4.2 Research Questions

RQ1: How do *overlap ratio* and *background noise* affect ASR accuracy under controlled synthetic conditions?

RQ2: How do publicly available ASR models compare across different overlap and noise operating points?

RQ3: To what extent do trends observed on controlled *synthetic mixtures* transfer to real conversational recordings with natural overlap?

RQ4: Does the proposed double-stream serialisation metric provide a more suitable evaluation than ordinary WER when a two-speaker reference is compared with a single-stream hypothesis?

The final question is accompanied by the following working hypothesis:

$$d_i = M_{\mathrm{DSS},i} - M_{\mathrm{WER},i}, \tag{4.1}$$

where $M_{\mathrm{DSS},i}$ and $M_{\mathrm{WER},i}$ are the DSS-WER and ordinary WER for the i-th clip, is not significantly different from zero across clips.

$$H_0 : \ \mathbb{E}[d_i] \geq 0 \qquad \text{against} \qquad H_1 : \ \mathbb{E}[d_i] < 0. \tag{4.2}$$

Rejection of H_0 therefore indicates that the proposed metric yields significantly lower scores than standard WER on average.

H_0 is expected to hold in the case of non-overlapping speech, where the single-stream hypothesis can be directly compared to the single-speaker reference. However, in the case of overlapping speech, H_1 should hold because wer assumes a single reference stream and will penalise interleaving errors that should not be penalised while DSS-WER can recognise the interleaving nature and therefore yield a lower score.

This hypothesis is not intended to imply that DSS-WER is universally superior to WER. This is specific to the evaluation problem studied in this dissertation, where a single-stream hypothesis is compared against a two-stream reference in which two speaker may overlap and transcripts may be interleaved.

4.3 Design Principles

The overall design principle is a repeated measures benchmarking framework. Each synthetic clip is transcribed by the same set of ASR models, and the model comparisons are paired at clip level. Within the synthetic benchmark, acoustic variants are also paired as the generation process first generate a clean base clip and then derives multiple SNR and noise-category variants from the same base. This reduces the variance, makes difference between conditions more attributable to corresponding factors and makes result more interpretable.

Two consideration motivated the choice of experimental factors. First, overlap ratio is treated as the primary factor of interest because it is the central phenomenon of interest in this dissertation and because overlapping speech is a common and challenging condition for ASR. Second, noise is crossed compared with overlap because both noise and overlap often co-occur in real-world conditions.

4.4 Track 1: Controlled Synthetic Benchmark

4.4.1 Speech and Noise Sources

The synthetic mixtures sample clean speech from the source root declared in `speech_roots.txt`, which includes:

- **LibriSpeech train-clean-100** LibriSpeech is a corpus of approximately 100 hours of 16kHz read English speech, prepared by Vassil Panayotov with the assistance of Daniel Povey. The data is derived from read audiobooks from the LibriVox project, and has been carefully segmented and aligned. [33]
- **CSTR VCTK Corpus**: a large-scale speaker recognition dataset containing speech from thousands of speakers, suitable for creating synthetic mixtures with varying overlap and SNR [34].

These datasets provided abundant clean speech with diverse speakers and accurate transcripts. The mixture generator scans these roots for `.wav` or `.flac` files and their corresponding transcripts, and converts multi-channel recordings to mono by channel averaging and resampling to 16kHz if necessary. The background noise is drawn from a local copy of the DEMAND dataset [35]. The DEMAND (Diverse Environments Multichannel Acoustic Noise Database) aims to provide a set of recordings that allow testing of algorithms using real-world noise in a variety of settings. The noise is categorised into three types: domestic (i.e., washroom, kitchen, living room), public (i.e., subway station, office cafeteria, university restaurant) and transportation (i.e., subway, public bus, private passenger vehicle), denoted as D, P, and T respectively. These noise types are broad categories that represent common real-world listening environments where ASR are often used, and small enough to support a manageable clean factorial analysis.

4.4.2 Synthetic Mixture Generation

The mixture generator takes a target overlap ratio, SNR level, noise type, speaker count and target duration as inputs. For the experiment here, the random seed is fixed to 1234, the target duration is set to 60s, speaker count is set to 2 in the configuration for reproducibility and isolation of the effect of OVR, SNR and noise type.

For each base mixture, two speakers are sampled without replacement from the pool of speakers. Utterances are then drawn from each speaker alternatively until the total duration exceed the required duration to realise the target final duration after overlapping. Each utterance also contributed its transcript in which the speaker ID, start and end timestamps are also preserved in a form of

$$(\text{speaker}, \text{text}, t_{\text{start}}, t_{\text{end}}). \tag{4.3}$$

This makes the manifest enable the calculation of any speaker-attributed or time-constrainted metrics in the future.

4.4.3 Noise Addition and SNR Control

The base mixture is then used to derive multiple variants with different noise types and SNR levels. The noise is rescaled by a factor α to achieve the target SNR level, as described in Equation (3.10). The final mixture is then

$$x_{\text{final}}(t) = x_{\text{speech}}(t) + \alpha x_{\text{noise}}(t) \tag{4.4}$$

The same clean base mixture is reused across all SNR and noise-category variants, which makes the variants paired and reduce variance. When SNR is `null` in the configuration, no noise is added and the base mixture is function as clean control condition.

4.4.4 Experimental Conditions and Corpus Size

Table 4.1: Experimental factors and planned levels.

Factor	**Levels**
Overlap ratios (OVRs)	$0\%, 14\%, 20\%, 40\%$
Max concurrent speakers (S)	2
SNR (dB)	clean, 7.4, 0, -5
Noise type	domestic, public, transportation
Audio duration	Synthetic: ≈60s, Real-world: long-form meetings
OVR std ratio	0.1

Table 4.2: Synthetic corpus size in the current manifest.

Target overlap ratio	**Base mixtures**	**Derived variants per base**	**Total clips**
0.00	100	12	1200
0.14	100	12	1200
0.20	100	12	1200
0.40	100	12	1200
Total			4800

Table 4.1 documents the levels of the main experimental factors. In the implementation, all conditions use 100 base mixtures. Each base mixture is expanded across all factors SNR × noise type, yielding a total of 4800 corpus summarised in Table 4.2.

Overlap Ratios Choice. The overlap ratios are chosen to represent a range of realistic and challenging conditions for ASR:

- 0% is chosen to be the control where it represent completely clean audio in which there is no overlapping.
- 14% is chosen to represent the mean overlap ratio for meetings. The percentage of overlapped words (excluding backchannels like "uh-huh") in meetings is 14.1% [36].
- 20% is chosen to represent the maximum overlap ratio in a natural meeting or casual dinner parties [3] [12].
- 40% is chosen as it is supported by the LibriMix benchmark, which chooses 40% as the maximum overlap ratio. This is chosen to stress-test under high-overlap conditions [3].

Maximum Concurrent Speakers Choice. Speaker count is chosen to be 2 for practicality and to isolate the effect of overlap and noise.

Signal-to-noise Ratio Choice. The SNR levels are chosen to represent a range of realistic and challenging noise conditions:

- *clean* is chosen to be the control where it represent clean background with no noise.
- 7.4dB is chosen to represent the noisy environment as it is the mean SNR for noisy environment [37].
- 0db is chosen to represent environment where the noise and speech are equally loud, which is a challenging but realistic condition in very noisy environments.
- -5db is chosen to represent environment where the noise is louder than speech in which even human listeners struggle to understand speech, providing a stress-test condition for ASR.

Noise Type Choice. Domestic, public, and transportation are chosen as they represent different real-world scenarios with varying noise characteristics:

- **Domestic noise** includes kitchen noise when food is being prepared, living room noise and washing room with front-loading washing machine. These noises are commonly found in home environments that a home-based voice assistant would encounter.
- **Public noise** includes noise from main transfer area of subway station, office cafeteria, and university restaurants during lunchtime. These noise represent common public environments where ASR like mobile devices may be used in.
- **Transportation noise** includes noises in subway, public bus and private passenger vehicle. These noises represent common transportation environments where ASR like mobile devices may be used in.

4.5 Track 2: Real-world Data Transferability Evaluation

The real-world data evaluation is intentionally narrower than the synthetic benchmark. Its purpose is to test whether the trends of the findings in synthetic mixtures survives when overlap is natural rather generated.

The implemented real-data scripts is extracted from CHiME-6 evaluation scripts [2], specifically the `S01` session. It is a English multi-channel far field meeting data used in the CHiME-6

Challenge. It is derived from CHiME-5 by fixing some array synchronization errors. It uses the 2 hours long audio and divides it into 100 segments of approximately 60s each by the following rules:

- each clip must be at least 60 s long;
- clip boundaries must lie at utterance boundaries where no other utterance is active at the boundary time;
- only utterances fully contained within the selected time window are kept in the reference.

These design choice avoid clipping utterances mid-sentence, perserving the accuracy of the reference. The 60s duration is such that the duration is similar to that of synthetic mixtures. They mean that the resulting clips are not fully representative of the natural distribution of overlap in the original recording, but that limitation is necessary to enable a clean comparison.

4.6 ASR Models Under Evaluation

Although the framework is designed to be extensible to any ASR model, the evaluation will focus on the following publicly available models that represent a range of model architectures:

- **Faster Whisper** (OpenAI): Faster-whisper is a reimplementation of OpenAI's Whisper model using CTranslate2, which is a fast inference engine for Transformer models. [38]
- **WhisperX** (OpenAI): WhisperX is a ASR model trained on a large dataset of diverse audio [39].
- **Wav2Vec2-Large-960h** (Meta AI): The large model pretrained and fine-tuned on 960 hours of Librispeech on 16kHz sampled speech audio. When using the model make sure that your speech input is also sampled at 16Khz. [40]
- **parakeet-tdt-0.6b-v3** (NVIDIA): parakeet-tdt-0.6b-v3 is a 600-million-parameter multilingual automatic speech recognition (ASR) model designed for high-throughput speech-to-text transcription. [41]

These models are chosen as representation of the mainstream ASR architectures. Faster Whisper and WhisperX are both based on the Whisper architecture, which is a encoder-decoder architecture trained on a large dataset. Wav2Vec-2.0 is an example of self-supervised encoders while Parakeet is an example of the conformer and transducer architecture. The evaluation of these models will provide insights into how different ASR architectures perform under overlapping and noisy conditions, and whether certain architectures are more robust in specific conditions.

4.7 Scoring Metrics

4.7.1 Reference and Hypothesis Structure

For synthetic data, the reference transcripts are loaded from the manifest and reduced to $(speaker, text)$ pairs (removing timestamps) as not all ASR models provide timed output. Depending on the model, the hypothesis may be a single stream of text or as multiple segments. This structure determines what metrics can be calculated as some metrics assume a segmented output while some do not. The normalisation function used in scoring is applied to both reference and hypothesis. It strips punctuation and converts text to lowercase. This is to isolate the evaluation to the core lexical robustness under overlap and noise rather than punctuation and casing.

4.7.2 Metrics

Three metrics are used in the synthetic scoring pipeline:

- **Word Error Rate (WER)**: The standard conventional metric for ASR evaluation, computed after concatenating all segments of reference and hypothesis by chronological order
- **Optimal Reference Combination WER (ORC-WER)**: Measures the minimum WER for multi-speaker scenarios across any mapping of hypothesis segements to reference utterances.
- **Multi-Reference Serialisation WER (DSS-WER)**: Proposed metric that computes the minimum edit distance between double reference chains and single hypothesis, allowing any interleaving merge of the two chains.

The DSS-WER implementation is based on the beam-search approximation algorithm in Section 3.6.3 with hyperparameters set to beam width of 64, lookahead of 16, heuristic weight of 0.4, max expansion 220000, max beam width of 128 and with adpative beam size and stratified beam pruning.

4.7.3 Metric Application Rules

The choice of metrics are based on the following rules:

- For single-stream hypothesis output, WER and DSS-WER are used.
- For segmented hypothesis output, all three WER, ORC-WER and DSS-WER are used.

4.8 Experimental Unit and Comparison Strategy

The fundamental unit of comparison is the (clip, model) pair with related metadata and transcript. For synthetic data, the comparison includes:

- comparison across models are paired at clip level as all models are evaluated on the same set of clips;
- comparisons across SNR values are paired becauses noisy variants are derived from the same base mixture;

This pairing structure supports stronger inference than purely between-subjects design. When a model performs different in lower SNR or higher overlap, the degradation or improvement can be attributed more confidently to the change in condition rather than the change in lexical content or other confounding factors.

4.9 Track 3: Hypothesis Testing of DSS-WER

This track evaluates whether the proposed Double-Stream Serialisation WER (DSS-WER) approximation algorithm differs systematically from standard WER, and whether it occurs in the expected settings and direction. Since DSS-WER is designed to relax the strict serialisation assumption of WER, the primary questions are whether DSS-WER yields lower scores than WER in overlapped setting while remaining similar in non-overlapped setting.

Let

$$d_i = M_{\mathrm{DSS},i} - M_{\mathrm{WER},i}, \tag{4.5}$$

where $M_{\mathrm{DSS},i}$ and $M_{\mathrm{WER},i}$ are the DSS-WER approximation and ordinary WER for the i-th clip. A negative value of d_i will therefore indicate that DSS-WER yields a lower score than WER on the i-th clip.

4.9.1 Primary Hypothesis

The main hypothesis is that examines whether DSS-WER approximation is lower than WER on average:

$$H_0 : \ \mathbb{E}[d_i] \geq 0 \qquad \text{against} \qquad H_1 : \ \mathbb{E}[d_i] < 0. \tag{4.6}$$

Rejection of H_0 therefore indicates that the proposed metric yields significantly lower scores than WER over the set of clips.

4.9.2 Condition-Specific Hypotheses

To further assess whether the expected advantage of DSS-WER approximation is specific to the overlapping condition, the test is repeated separately for different overlap ratio.

Non-overlapping clips. For clips with no overlap, DSS-WER is expected to behave similarly to WER. Therefore, the hypothesis:

$$H_0 : \ \mathbb{E}[d_i] = 0 \qquad \text{against} \qquad H_1 : \ \mathbb{E}[d_i] \neq 0. \tag{4.7}$$

Large d_i will suggest that DSS-WER approximation alters the evaluation even when interleaving is not a factor.

Overlapping clips. For clips with overlap, DSS-WER is expected to be lower than standard WER, since it permits interleaving. Therefore the hypothesis is the same as the primary hypothesis:

$$H_0 : \ \mathbb{E}[d_i] \geq 0 \qquad \text{against} \qquad H_1 : \ \mathbb{E}[d_i] < 0. \tag{4.8}$$

4.9.3 Test Procedure

Since metrics are calculated at clip level, the comparison is paired where hypothesis testing is conducted on paired differences d_i. A one-sided Monte Carlo permutation test may be used by randomly flipping the signs of the paired differences to generate the null distribution of the test statistic $\bar{d}$. The observed mean

$$T_{\text{obs}} = \bar{d} = \frac{1}{N} \sum_{i=1}^{N} d_i \tag{4.9}$$

is then compared against the null distribution generated by random sign flipping, for $j = 1, ..., N$ permutations:

$$T_{\text{null}}^{(j)} = \frac{1}{N} \sum_{i=1}^{N} s_i^{(j)} d_i, \qquad s_i^{(j)} \in \{-1, 1\} \text{ with equal probability.} \tag{4.10}$$

The p-value is calculated as

$$p = \frac{1}{N+1} \left(1 + \sum_{j=1}^{N} 1[T_{\text{null}}^{(j)} \geq T_{\text{obs}}] \right), \tag{4.11}$$

where $1[\cdot]$ is the indicator function. The confidence interval of the mean difference can also be calculated by resampling the paired differences with replacement, calculating the mean difference for each resample and take the 2.5 and 97.5 percentiles of the resampled mean differences as the bounds of the 95% confidence interval.

4.9.4 Interpretation

This analysis does not intend to claim that DSS-WER is universally superior to WER. It test whether DSS-WER yields lower scores than WER in the double-stream reference single-stream hypothesis setting, where DSS-WER is designed to relax the interleaving restriction of WER.

4.10 Reproducibility

The implemented framework traces every results (including synthetic mixture generation, ASR inferencing and metric calculation) to its generating condition. The manifest stores clip identifiers, audio paths, source-file provenance, noise-file provenance, target overlap ratio, realised overlap ratio, SNR, and noise category. Model outputs are written to JSON keyed by clip ID, and evaluation outputs are stored both per clip and per model. Combined with the fixed random seed in the configuration, this provides a reproducible record of how each reported score was obtained.

4.11 Scope and Limitations

The experimental design combines precisely controlled synthetic benchmarking with a narrower but meaningful real-world data evaluation. It separates overlap and noise as explicit variables, preserving the paired comparisons across conditions.

However, the limitation is clear. They are as follows:

- the comparative system only cover four ASR models
- Not every model is run on every clip due to runtime and implementation constraints, which may introduce bias in the comparison.
- ORC-WER is only calculated for segmented output (faster-whisper and WhisperX) as not every model provides segment-level output
- Real-world evaluation only cover one session in CHiME-6 rather than a balanced sampling of real-world conditions

However, these limitations do not invalidate the design. They define the scope of the dissertation and the interpretation of the results.

Chapter 5

Statistical Analysis

5.1 Analytical Aims and Data Provenance

This chapter analyses the outcomes produced by the architectural design in Chapter 3 and the experimental design in Chapter 4. The primary aim is to answer the research questions outlined in Section 4.2. The analysis will focus on the following key questions:

RQ1: How do *overlap ratio* and *background noise* affect ASR accuracy under controlled synthetic conditions?

RQ2: How do publicly available ASR models compare across different overlap and noise operating points?

RQ3: To what extent do trends observed on controlled *synthetic mixtures* transfer to real conversational recordings with natural overlap?

RQ4: Does the proposed double-stream serialisation metric provide a more suitable evaluation than ordinary WER when a two-speaker reference is compared with a single-stream hypothesis?

The analysis uses the top-level result files generated by the evaluation pipelines described in Chapter 3, which contain the WER, DSS-WER, ORC-WER and other relevant metrics for each ASR model across all experimental conditions.

Two analysis scopes are kept separate:

1. A 700 clips for each of the 4 ASR models. They are a cross-model subset of the 4800 clips generated for the synthetic evaluation, defined as

$$\begin{aligned}\mathcal{S}_{\text{model}} = \{&\text{SNR} = 7.4,\ \text{noise} = T,\ \text{OVR} \in \{0.00, 0.14, 0.20, 0.40\}\} \\ &\cup \{\text{OVR} = 0.14,\ \text{noise} = T,\ \text{SNR} \in \{\text{clean}, 7.4, 0, -5\}\}.\end{aligned} \tag{5.1}$$

   Each combination of factors contains 100 clips, for a total of 700 per model, as shown in Figure 5.1.

2. the full set of 4800 clips for the `wav2vec-2.0` model, which is the only model evaluated across all conditions.

The first scope allows direct comparison of the 4 ASR models across the same set of clips. The reason for not using the full 4800 clips for all models is due to runtime and computational constraints. This is suitable for fair model ranking but should not be used to make claims about the effects of SNR-overlap combinations across all models. The second scope allows a more comprehensive analysis of the effects of overlap and noise on the ASR performance of `wav2vec-2.0`, which has the best balance of performance and runtime among the models evaluated.

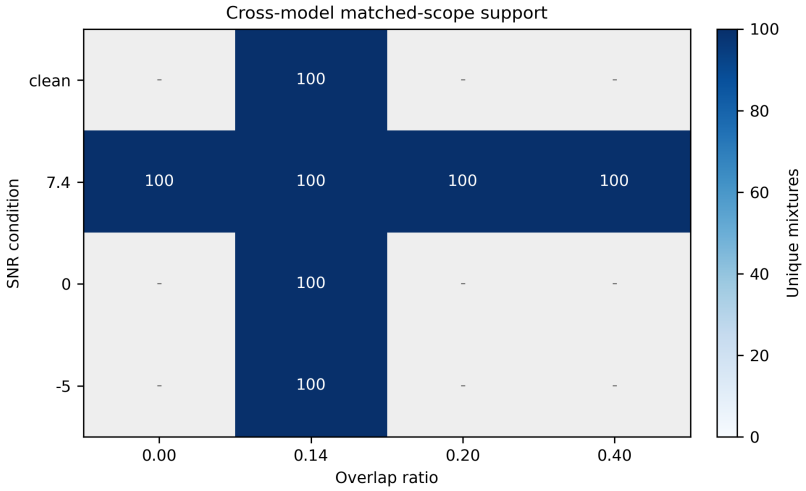


Figure 5.1: Unique mixture support in the cross-model comparison scope. Each included SNR–OVR condition contains 100 mixtures; grey cells are outside the planned matched subset.

5.2 Variable Selection

5.2.1 Dependent Variables

The primary dependent variable is the clip-level Word Error Rate (WER). The reason for choosing WER is that it is the de facto metric for ASR evaluation [42] and is available for all four models, including segmented and non-segmented hypotheses.

ORC-WER and DSS-WER are also available, where ORC-WER is only available for the segmented hypothesis (from `faster-whisper` and `WhisperX`) and DSS-WER is available for all models. They act as secondary metrics to provide additional insights into the performance. They are valuable as they are designed to handle different situations. ORC-WER is designed to represent the speaker-agnostic lexical accuracy of the segmented hypothesis. In contrast, DSS-WER is designed to evaluate the specific case of comparing a two-stream reference with a single-stream hypothesis, which is the case for this evaluation. However, since it is proposed in this dissertation, it is important to evaluate its behaviour and suitability before applying it as the primary metric. Therefore, WER is the most suitable and thus chosen as the primary metric for model comparison. At the same time, DSS-WER will be tested for its suitability and insights as a secondary metric.

5.2.2 Explanatory Variables

The main explanatory variables are signal-to-noise ratio (SNR), overlap ratio (OVR), noise type and ASR model. SNR and OVR are the two main factors of interest in ASR evaluation: noise and overlap. Overlap is the central phenomenon being investigated in this dissertation. Noise type is included as it might interact with the effects of SNR and OVR. The ASR model is included because it is important to understand how different models perform under the same conditions and whether the effects of SNR and OVR differ across models.

The speaker count is held constant at 2 for all clips. This makes OVR more interpretable as a controlled measure of two-speaker overlap rather than a more complex interaction between overlap and speaker count. The clip duration is also set to 60 seconds, which supports comparability across clips and models, providing enough speech content while keeping inference time manageable.

5.3 Statistical Methods

All reported results use the clip as the unit of analysis. The design is a factorial design with repeated measures, where the same base audio is used across different SNR and noise type conditions; cross-model analysis uses the same clip IDs system across models. For this reason, the analysis will focus on matched condition summaries and comparisons.

WER is non-negative and often skewed, so the mean is reported as the primary summary statistic. Still, the median and standard deviation are also reported to provide a more complete picture.

5.4 Overview of the Evaluation Results

5.4.1 Scope 1: Cross-Model Comparison

Table 5.1: Cross-model availability of WER, ORC-WER and DSS-WER.

Model	WER	ORC-WER	DSS-WER
faster-whisper	yes	yes	yes
WhisperX	yes	yes	yes
wav2vec-2.0	yes	no	yes
Parakeet	yes	no	yes

Table 5.2: WER-first model comparison on the matched 700-clip subset.

Model	**Mean WER**	**Median WER**	**SD**	**Mean ORC**	**Mean DSS**
faster-whisper	0.159	0.123	0.128	0.604	0.156
WhisperX	0.181	0.139	0.150	0.584	0.179
wav2vec-2.0	0.228	0.205	0.133	-	0.211
Parakeet	0.503	0.525	0.208	-	0.515

Section 5.4.1 shows the availability of each metric across the different models. ORC-WER is not available for `wav2vec-2.0` and `Parakeet`, as OCR-WER is designed for segmented hypotheses, while wav2vec-2.0 and Parakeet output a single continuous sequence. Table 5.2 shows the mean, median and standard deviation of WER for each of the models. The results show that `faster-whisper` has the lowest mean WER, while `Parakeet` has the highest mean WER. Figure 5.2 shows the distribution of WER across the four models. The results show that `faster-whisper` and `WhisperX` are more right-skewed than the other two, which might be due to their lower WER values.

5.4.2 Inference Time

The timing benchmark uses 100 randomly selected clips from the same synthetic corpus. Time per The clip is computed as the total processing time divided by the number of clips. RTFx is computed as the total audio duration divided by total processing time; a larger value, therefore, indicates faster-than-real-time processing. In Figure 5.3a and Table 5.3, `wav2vec-2.0` is the fastest model with a mean inference time of 0.754 seconds per clip, while `faster-whisper` is the slowest with a mean inference time of 27.736 seconds per clip. The RTFx values indicate the length of audio that can be processed in one second of real time, with `wav2vec-2.0` achieving the highest RTFx of 88.951. As shown in Figure 5.3b, the combination of relatively low WER and

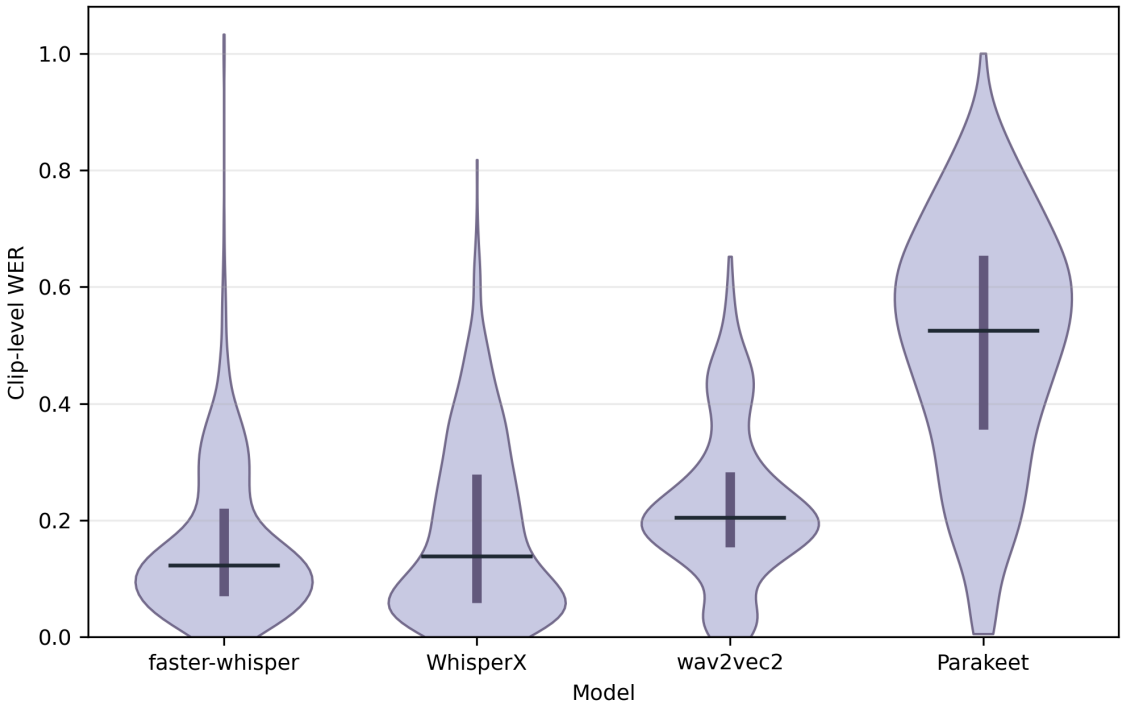


Figure 5.2: Violin plot of WER across different ASR models.

high RTFx makes `wav2vec-2.0` the candidate for the second scope of analysis, which focuses on the effects of SNR and OVR on a single model.

Table 5.3: Inference time and real-time factor on the 100-clip timing benchmark.

Model	**Clips**	**Time per clip (s)**	**RTFx**
wav2vec-2.0	100	0.754	88.951
Parakeet	100	1.124	59.663
WhisperX	100	7.276	9.220
faster-whisper	100	27.736	2.419

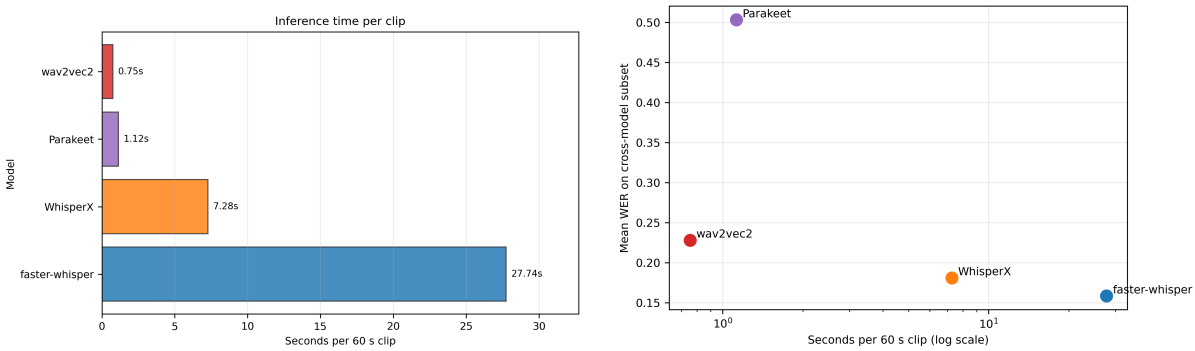
5.5 Scope 2: Full Condition Analysis of wav2vec-2.0

There are 48 conditions in total.

$$\mathcal{S}_{\text{full}} = \{\text{OVR} \in \{0.00, 0.14, 0.20, 0.40\},\ \text{noise} \in \{D, P, T\},\ \text{SNR} \in \{\text{clean}, 7.4, 0, -5\}\}, \quad (5.2)$$

where each condition contains 100 clips, the heatmap shows a clear trend of increasing WER with higher overlap and lower SNR (noisier conditions).

Figure 5.4 shows the WER heatmap for `wav2vec-2.0` across different SNR and overlap conditions. The results show a clear trend of increasing WER with higher overlap and lower SNR (noisier conditions). In Figure 5.5, the WER is compared across the three noise types. The results show that public noise (P) has the highest WER, followed by domestic noise (D) and transportation noise (T). There is a marginal difference between the noise types, with (P) being almost 0.3 WER higher than the lowest noise type (T). This suggests that public noise might be more challenging for ASR, owing to its potentially more irregular and unpredictable nature,



(a) Inference time per clip.

(b) WER–latency trade-off.

Figure 5.3: Inference efficiency across ASR models. The left panel reports mean inference time per approximately 60-second clip, while the right panel compares timing with matched-subset WER.

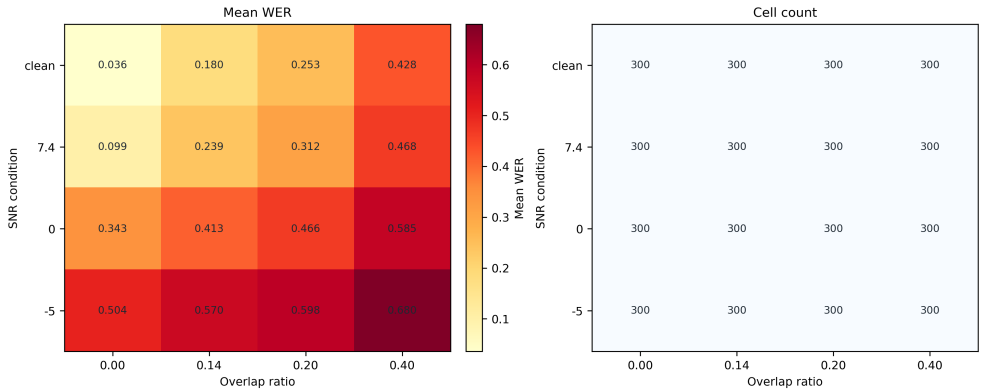


Figure 5.4: WER heatmap for wav2vec-2.0 across different SNR and overlap conditions.

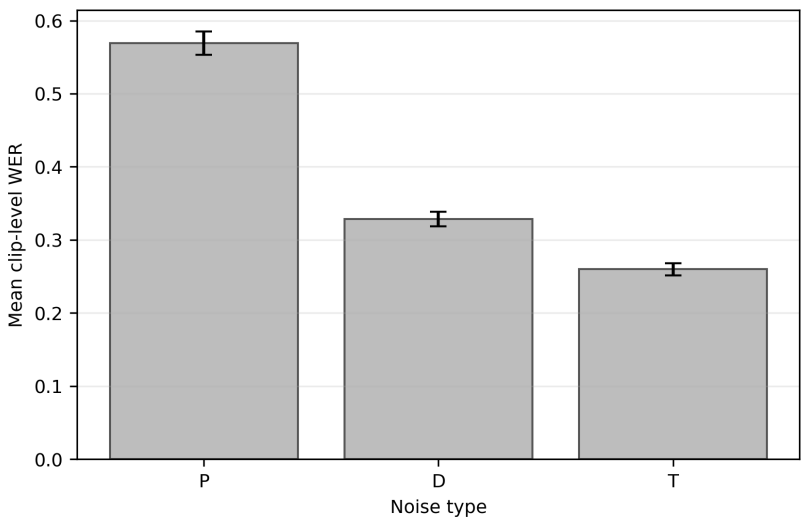
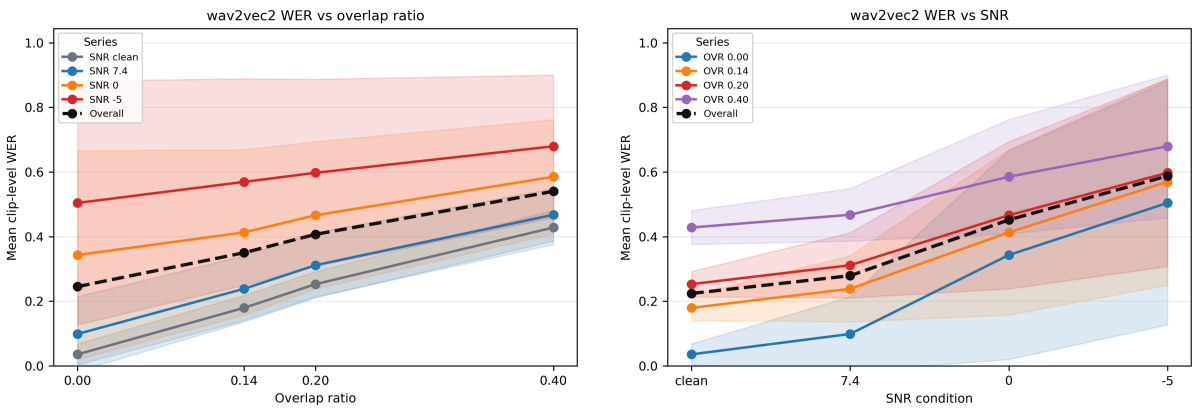


Figure 5.5: wav2vec2 mean WER by DEMAND noise category. Error bars show approximate 95% confidence intervals.

compared to the other two, while transportation noise might be the least challenging due to its more consistent and predictable acoustic patterns.

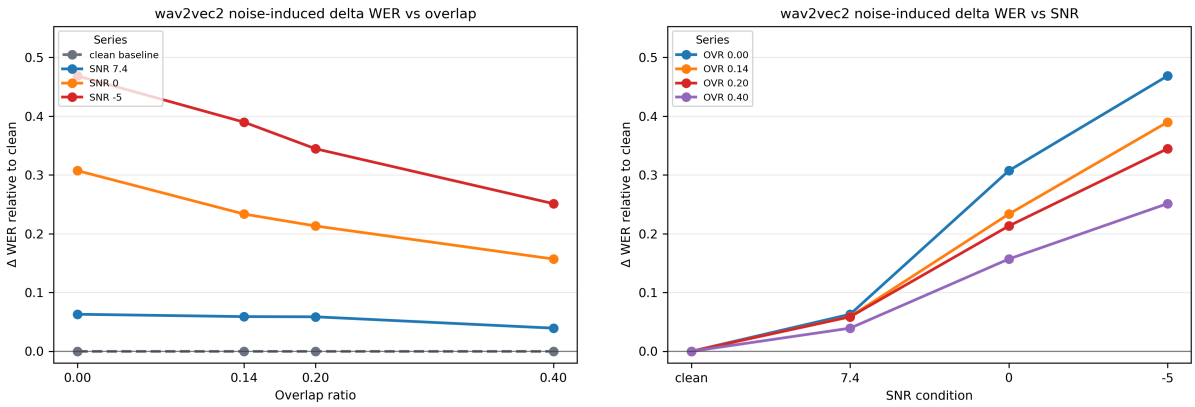
This section went through the coverage and reliability of the evaluation outputs. The following sections analyse the results by research question, starting with the effects of the overlap ratio and signal-to-noise ratio.

5.6 RQ1: Effects of Overlap Ratio and Signal-to-Noise Ratio



(a) WER against overlap ratio, stratified by SNR. (b) WER against SNR condition, stratified by OVR.

Figure 5.6: Mean wav2vec-2.0 WER trends across the full synthetic sweep. Each panel includes condition-specific lines and an overall pooled mean.



(a) Delta WER against OVR, stratified by SNR. (b) Delta WER against SNR, stratified by OVR.

Figure 5.7: Delta WER for wav2vec-2.0. Delta (a) is computed relative to the clean condition at the same overlap ratio, while Delta (b) is computed relative to the zero overlap condition at the same SNR.

5.6.1 Effect of Overlap Ratio

Figure 5.6a shows the general trends of WER against the overlap ratio. The results show a notable increase in WER as the overlap ratio increases.

Figure 5.7b shows the delta WER $\Delta_{\text{WER, 0\% OVR}}$ against SNR, where

$$\Delta_{\text{WER, 0\% OVR}} = \text{WER}_{\text{SNR,0\%}} - \text{WER}_{\text{SNR,OVR}}. \tag{5.3}$$

This isolates the effect of overlap as SNR changes. The results show that the $\Delta_{\text{WER, 0\% OVR}}$ increases as the SNR decreases, which suggests that the effect of overlap is more substantial when SNR is low (noisier conditions). The $\Delta_{\text{WER, 0\% OVR}}$ diverges across different overlap ratio conditions as the SNR decreases, which matches the observation from Figure 5.7a that the effect of overlap becomes more notable as the SNR decreases.

5.6.2 Effect of Signal-to-Noise Ratio

Figure 5.6b shows the general trends of WER against SNR. The results show a considerable increase in WER as the SNR decreases (i.e., the signal becomes noisier). Figure 5.7a shows the delta WER $\Delta_{\text{WER, clean SNR}}$ against overlap ratio, where

$$\Delta_{\text{WER, clean SNR}} = \text{WER}_{\text{SNR,OVR}} - \text{WER}_{\text{clean,OVR}}. \tag{5.4}$$

This isolates the effect of noise as the overlap ratio changes. The results show that the $\Delta_{\text{WER, clean SNR}}$ decreases as the overlap ratio increases, which suggests that the effect of noise is more substantial when the overlap ratio is low, and decreases as the overlap ratio increases. The $\Delta_{\text{WER, clean SNR}}$ starts to converge across different SNR conditions as the overlap ratio increases, which suggests that the effect of noise starts to become dominated by the effect of overlap as the overlap ratio increases.

Figure 5.7b shows the delta WER $\Delta_{\text{WER, 0\% OVR}}$ against SNR, where

$$\Delta_{\text{WER, 0\% OVR}} = \text{WER}_{\text{SNR,0\%}} - \text{WER}_{\text{SNR,OVR}}. \tag{5.5}$$

This isolates the effect of overlap as SNR changes. The results show that the $\Delta_{\text{WER, 0\% OVR}}$ increases as the SNR decreases, which suggests that the effect of overlap is more substantial when SNR is low (noisier conditions). The $\Delta_{\text{WER, 0\% OVR}}$ diverges across different overlap ratio conditions as the SNR decreases, which matches the observation from Figure 5.7a that the effect of overlap becomes more notable as the SNR decreases.

To summarise the trend as a continuous estimating equation, SNR is converted to an amplitude noise ratio

$$z(\text{SNR}) = \begin{cases} 0, & \text{clean } (\text{SNR} = \infty), \\ 10^{-\text{SNR}_{\text{dB}}/20}, & \text{otherwise}, \end{cases} \tag{5.6}$$

changing SNR from logarithmic to linear scale, making SNR a continuous variable. Fitting a least-squares surface to the full `wav2vec-2.0` scope gives

$$\widehat{\text{WER}} = 0.029 + 1.001\,\text{OVR} + 0.274\,z - 0.328\,(\text{OVR} \times z). \tag{5.7}$$

The interaction term is retained because the noise penalty is not constant across overlap ratios. This equation should be interpreted as an estimate of the mean WER averaged over noise type: it gives a strong fit to the 16 SNR–OVR condition means ($R^2 = 0.986$, MAE = 0.017 WER), where

$$\text{R}^2 = 1 - \frac{\sum_{i \in \{\text{SNR}\times\text{OVR}\}} (\text{WER}_i - \widehat{\text{WER}}_i)^2}{\sum_{i \in \{\text{SNR}\times\text{OVR}\}} (\text{WER}_i - \overline{\text{WER}})^2}, \tag{5.8}$$

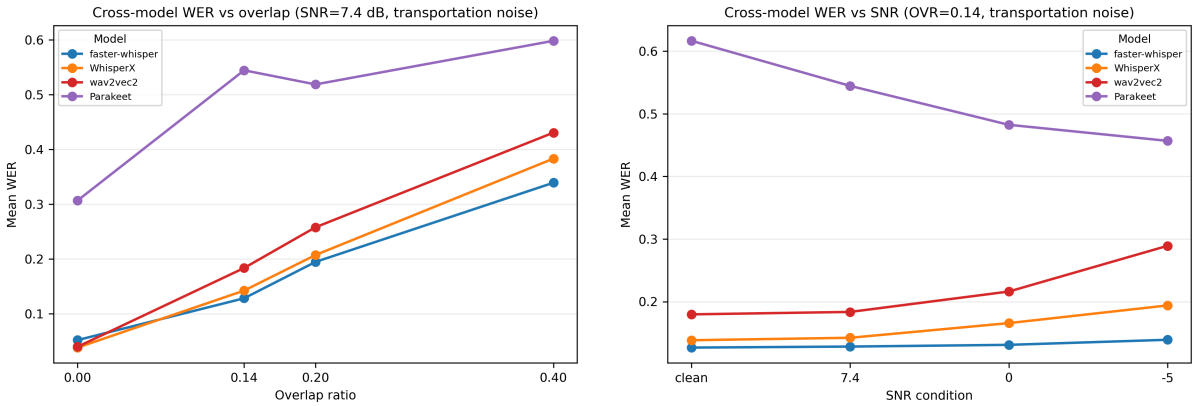
$$\text{MAE} = \frac{1}{16} \sum_{i \in \{\text{SNR}\times\text{OVR}\}} |\text{WER}_i - \widehat{\text{WER}}_i|, \tag{5.9}$$

where R^2 is the coefficient of determination, MAE is the mean absolute error, and $\overline{\text{WER}}$ is the mean WER across the 16 conditions. The clip-level fit is lower ($R^2 = 0.433$, MAE = 0.148 WER) because individual clips vary substantially in lexical content and acoustic difficulty. The result shows that the OVR dominates the WER increase as it has a larger coefficient than the noise term. However, the negative interaction term indicates an antagonistic relationship between noise and overlap, suggesting the marginal effect of one degradation factor is smaller when the other is already severe.

5.6.3 Summary

The results show a positive relationship between WER and the overlap ratio, and between WER and noise (noisy conditions have higher WER than the clean condition). The effect of the overlap ratio is more substantial in noisier conditions, while the effect of noise is stronger in lower overlap conditions. The fitted equation suggests that the effect of overlap ratio is more significant than that of noise, but they interact in an antagonistic manner, reducing each other's impact on WER.

5.7 RQ2: Comparison of ASR Models



(a) WER against overlap ratio, stratified by SNR. (b) WER against SNR condition, stratified by OVR.

Figure 5.8: Mean WER trends across the cross-model subset. Each panel includes condition-specific lines and an overall pooled mean.

5.7.1 Overall Comparison

Table 5.2 records the WER, ORC-WER and DSS-WER for the four ASR models on the matched 700-clip subset. The results show that `faster-whisper` has the lowest mean WER of 0.159, while `Parakeet` has the highest mean WER of 0.503, with a substantial gap between the second-highest model `wav2vec-2.0` of 0.228. The ORC-WER and DSS-WER values are generally consistent with the WER ranking, with `faster-whisper` having the lowest DSS-WER. However, the ORC-WER values for `faster-whisper` are higher than `WhisperX` even though `faster-whisper` has a lower WER. This suggests that `faster-whisper` might have a better lexical accuracy but worse segmentation accuracy than `WhisperX`.

Figure 5.8 shows the WER trends against overlap ratio and SNR for the four ASR models. Both show the same ranking of the models across conditions, with `faster-whisper` < `WhisperX` < `wav2vec-2.0` < `Parakeet`, which is consistent with the overall WER ranking in Table 5.2. The gap between the models remains relatively consistent as the overlap ratio increases, but the gap decreases as the SNR decreases. It is interesting to note that `Parakeet` 's performance is counterintuitively better as noise increases, which might be due to differences in the pretraining data and architecture. In addition, it has a dip in WER (slight increase in WER) from 0.14 OVR to 0.20 OVR. Neither of these behaviours is observed in the other three models, which might suggest that `Parakeet` and the architectural design (conformer and transducer-based architecture) are more resistant to noise. This is an interesting finding that warrants further experimentation and analysis, but it is not the main focus of this dissertation, so it will not be further investigated in this chapter.

5.7.2 Extreme Cases

This section analyses the worst-performing clip for each model.

Table 5.4: Distribution of high-WER outliers by model (IQR rule: WER $> Q_3 + 1.5\,\mathrm{IQR}$). Proportions are within each model's outlier set.

Model	$n_{\mathbf{out}}/n$	**Rate (%)**	**OVR (%)**				**SNR (%)**			
			0.00	**0.14**	**0.20**	**0.40**	**clean**	**7.4**	**0**	**-5**
faster-whisper	20/700	2.86	5.0	40.0	20.0	35.0	10.0	70.0	5.0	15.0
WhisperX	9/700	1.29	0.0	44.4	22.2	33.3	11.1	66.7	11.1	11.1
wav2vec-2.0	19/4800	0.40	31.6	21.1	31.6	15.8	0.0	0.0	10.5	89.5
Parakeet	0/700	0.00	–	–	–	–	–	–	–	–

As shown in Table 5.4, outliers are sparse across models, but when present, they are concentrated in specific operating points. The outliers are heavily concentrated in the SNR = 7.4 condition for `faster-whisper` and `WhisperX` with outliers, which is the second-best SNR condition. This is counterintuitive, as one would expect more outliers in the noisiest condition (SNR $= -5$). This might suggest that at an SNR of 7.4, the noise is more likely to trigger hallucinations in the models. It might be that quieter noise is more likely to be misinterpreted as speech, forcing the models to generate some words. In comparison, louder noise is more likely to be correctly identified as noise and thus less likely to trigger hallucination. This warrants further investigation, but it is outside the scope of this study. In contrast, the outliers for `wav2vec-2.0` are heavily concentrated in the noisiest condition (SNR $= -5$), which is more intuitive as the model might be more likely to miss or misrecognise speech in the noisiest condition, leading to higher WER. On the other hand, there are zero outliers for `Parakeet`, which is consistent with the fact that it has the highest mean WER and standard deviation, suggesting that the performance of `Parakeet` varies widely across clips but does not have extreme outliers (due to the larger IQR).

`faster-whisper`. The worst performing clip: Clip ID: `mix_0004361_0.40_2_7.4_T`; WER: 1.033; Condition: OVR = 0.40, SNR = 7.4, noise type = T.

> **Hypothesis (excerpt):** The project ... the director. Thank you for watching. If you like this video, please subscribe to the channel. ...

In the example above, after the relevant hypothesis is finished, the T̈hank you for watching. If you like this video, please subscribe to the channel.ïs repeated twice and then Ïf you like this video, please subscribe to the channel.ïs repeated 11 more times. It is worth noting that this specific repeating sentence or anything similar is not in any part of the audio. Also, every T̈hank you for watching.änd Ïf you like this video, please subscribe to the channel.äre output as standalone segments. After inspecting the audio of the clip, it is confirmed that there is no corresponding audio for these repeated sentences. So this sentence is purely hallucinatory. Hence, the hypothesis exhibits severe hallucinations and repetition, likely due to `faster-whisper` 's LLM-based decoding strategy, which is more prone to hallucinations, especially under difficult conditions. If the hallucinated tokens (all prefixing "Thank you for watching. If you like this video, please subscribe to the channel.") are removed, the WER would be reduced to 0.509, which is still a high WER but much more in line with the other results in the same condition. After inspecting the worst outliers for `faster-whisper` (3 worst being `mix_0004361_0.40_2_7.4_T`, `mix_0003509_0.20_2_7.4_T`, `mix_0001484_0.14_2_0_T`), it is found that they all contain severe hallucination. However, there is no clear pattern in terms of the condition of the clips, as they varied across different SNR and OVR conditions. Table 5.4 shows that the outliers show no clear

relationship in OVR but show a strong relationship with SNR, with 70% of the outliers in the SNR = 7.4 condition. This might be due to the reason mentioned above: quieter noise is more likely to be misrecognised as speech and thus trigger hallucinations.

`WhisperX`. The worst performing clip: Clip ID: `mix_0002729_0.20_2_7.4_T`; WER: 0.818; Condition: OVR = 0.20, SNR = 7.4, noise type = T.

> **Hypothesis (excerpt):** A week later ... need for him.Ër̈om that day forth, Duke was never referred to. ... From that day forth and equipped with ...

In this example, the sentence "From that day forth, Duke was never referred to." is repeated 18 times, but it came back to the correct sentence Ër̈om that day forth and equipped with ...ät the end. After inspecting the audio of this clip, it is confirmed that there is clear speech in the region where the hallucination occurs, indicating that the model starts repeating the last recognised sentence without any apparent trigger, while missing everything during that time. The hallucinated part is part of the same segment of the model's hypothesis output, unlike the hallucination in `faster-whisper`. This suggests that the hallucination might be due to the decoding strategy of `WhisperX`, where `faster-whisper` and `WhisperX` may use different strategies, leading to distinct hallucination patterns. If the hallucinated portion is removed, the WER would drop to 0.484. After inspecting the worst outliers (3 worst being `mix_0002729_0.20_2_7.4_T`, `mix_0004133_0.40_2_7.4_T`, `mix_0003917_0.40_2_7.4_T`), where they all contain hallucination. Similar to `faster-whisper`, there is no clear pattern in terms of the overlap ratio, but outliers are concentrated in the SNR = 7.4 condition, with 66.7% of the outliers.

`wav2vec-2.0`. The worst performing clip: Clip ID: `mix_0000454_0.00_2_-5_P`; WER: 1.0; Condition: OVR = 0.00, SNR = -5, noise type = P.

> **Hypothesis (excerpt):** O

In this example, the hypothesis is just a single character "O", which is likely due to the model completely missing the speech content in the clip, which is consistent with the fact that this clip is in the noisiest condition (SNR = -5). After inspecting the worst outliers, where the second worst clip is `mix_0000586_0.00_2_-5_P` and WER of 1.0 with

> **Hypothesis (excerpt):** MAASOO BERAE RAIOAAEAO OEMAASSA AOSA

and third worst clip is `mix_0000742_0.00_2_-5_P` and WER of 1.0 with

> **Hypothesis (excerpt):**EANOEAOAAIIOAOA

All of the worst outliers for `wav2vec-2.0` are in the noisiest condition (SNR = -5). After listening to the clips, the speech content was in the noisiest condition. It also suggests that wav2vec-2.0's LM component might play a lighter role in decoding or be less prone to hallucination, as the outliers are more likely due to missing speech content than to hallucination.

`Parakeet`. There is not outlier for `Parakeet` but the worst performing clip is: Clip ID: `mix_0001358_0.14_2_None_T`; WER: 1.0; Condition: OVR = 0.14, SNR = $None$, noise type = T.

> **Hypothesis (excerpt):**

In this example, the hypothesis is empty, which is counterintuitive as the clip is in a relatively easy condition with a clean condition. This might be an error in the decoding process of

Parakeet, as it has not happened with any other models. After inspecting the audio, there is no clear issue, so it is likely the model just failed to generate any output for this clip.

5.7.3 Inference Efficiency

Figure 5.3b shows the WER-latency trade-off across the four ASR models. The horizontal axis is the log scale of the mean inference time per clip, and the vertical axis is the mean WER on the matched 700-clip subset. The results show that wav2vec-2.0 offers the best trade-off, with a relatively low WER and the lowest inference time. Parakeet has the worst trade-off with the highest WER and the second-lowest inference time. faster-whisper has the lowest WER but the highest inference time, while WhisperX has a slightly higher WER but lower inference time. In Table 5.3, the wav2vec-2.0 has an RTFx of 89, which means it can process 89 times the length of audio in one second of processing time, which makes it most suitable for real-time applications among the four models. faster-whisper has an RTFx of 2.4, which makes it less suitable for real-time applications, but it has the lowest WER, which makes it a good model if accuracy is the primary concern and inference time is not a constraint. However, since all four models have an RTFx greater than 1, they can all process in real time. Hence, the choice of model depends on the specific requirements of the application, such as the acceptable WER and available computational resources.

5.7.4 Summary

The results show that, in terms of WER, faster-whisper has the best performance, while in terms of inference time, wav2vec-2.0 has the best performance. The WER trends across SNR and OVR are generally consistent and intuitive across the four models, with higher WER in higher OVR and lower SNR conditions. Overall, wav2vec-2.0 has the best accuracy-latency trade-off, while faster-whisper has the best accuracy but the worst latency. WhisperX is a balancing option between faster-whisper and wav2vec-2.0, with a slightly higher WER than faster-whisper but much lower inference time. Parakeet has the worst WER but a relatively low inference time, which makes it not the best choice for either accuracy or latency. In conclusion, the rank of the model will be as follows: In terms of WER: faster-whisper < WhisperX < wav2vec-2.0 < Parakeet; in terms of inference time: wav2vec-2.0 < Parakeet < WhisperX < faster-whisper; in terms of accuracy-latency trade-off: wav2vec-2.0 < WhisperX < faster-whisper < Parakeet.

5.8 RQ3: Transferability of ASR Performance Trends

To test the transferability of the synthetic benchmark results to real audio, the cross-model subset results are compared with those on 100 real CHiME-6 clips. The 100 real clips are segments of the longer CHiME-6 eval S01 recordings, which are 2 hours of real conversational audio recorded in a dinner party setting. The 100 clips are at least 60 seconds long and feature audio from multiple speakers. The WER and OCR-WER are computed for the 100 real clips.

Table 5.5: Synthetic-to-real transferability summary. Synthetic WER is computed on the matched 700-clip subset; real WER and real ORC-WER are computed on 100 CHiME-6 clips. Parentheses show standard deviation. ORC-WER is only available for segmented hypotheses.

Model	Synth. WER (std)	Real WER (std)	Real med.	Δ mean	Real ORC	Rank
faster-whisper	0.159 (0.128)	0.458 (0.182)	0.427	+0.299	0.797	1 → 3
WhisperX	0.181 (0.150)	0.436 (0.143)	0.413	+0.255	0.761	2 → 2
wav2vec-2.0	0.228 (0.133)	0.723 (0.123)	0.731	+0.495	–	3 → 4
Parakeet	0.503 (0.208)	0.389 (0.129)	0.357	-0.115	–	4 → 1

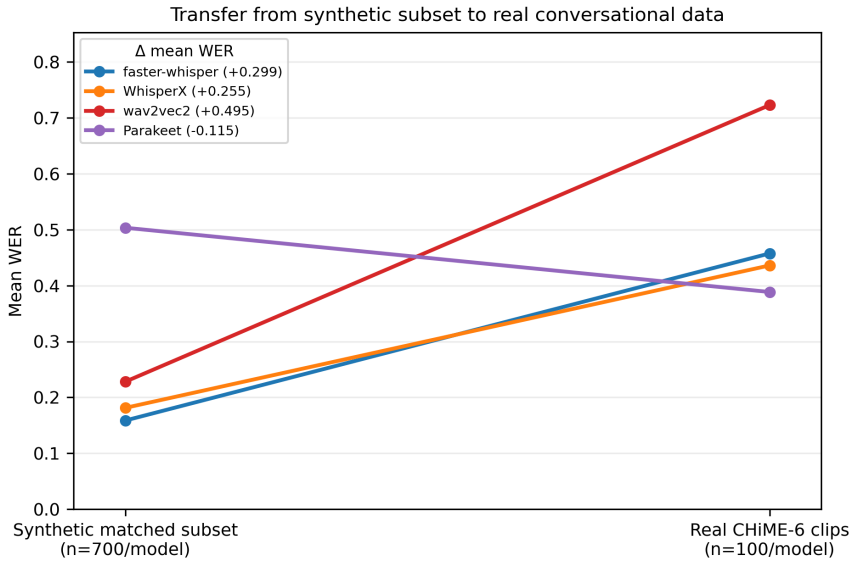


Figure 5.9: Change in mean WER from the matched synthetic 700-clip subset to the 100 real CHiME-6 clips. The legend reports the real-minus-synthetic change in mean WER.

Figure 5.9 and Table 5.5 show that the synthetic ranking only partially transfers to the real conversational setting. `WhisperX` and `faster-whisper` show a similar increase in WER from synthetic to real of about 0.4, with `faster-whisper` having a slightly higher increase. `wav2vec-2.0` has the highest degradation in WER of 0.495, while `Parakeet` has the largest reversal, where it shows a decrease in WER of about 0.1. In terms of ranking, `Parakeet` has become the best performing model in the real setting, while `faster-whisper` has dropped from first to third and `wav2vec-2.0` has dropped from third to fourth. In terms of ORC-WER, the two models with segmentation (`faster-whisper` and `WhisperX`) show a similar ORC-WER of around 0.8, which is much higher than the WER, suggesting that it is much more difficult for the models to segment the real audio correctly.

This suggests that the synthetic benchmarks are useful for understanding the relative effects of OVR and SNR, but have limited utility for ranking models in real conversational settings and thus should not be used as the sole basis for model selection in real-world applications. The real data includes additional information, such as natural speaker interaction, reverberation, and other acoustic factors (e.g., simultaneous conversations between two pairs of speakers) that are not fully captured in the synthetic data, which can significantly influence model performance. The results also suggest that `Parakeet` might be the most robust model in real conversational settings.

However, it is worth noting that the test set is relatively small, with only 100 clips, which may not fully reflect the actual performance. In addition, all the real clips are derived from the same recording, which might only represent a specific type of real conversational setting. Therefore, further testing on larger and more diverse datasets is needed to confirm the transferability of the synthetic dataset.

5.9 RQ4: Evaluation of DSS-WER

This section evaluates how DSS-WER behaves compared to ordinary WER, and whether it shows the expected behaviour based on the experimental design in Section 3.6.3 and Section 4.9. This analysis uses all successful synthetic results from all four ASR models. This means, 700 clips for `faster-whisper`, `WhisperX`, and `Parakeet`, and 4800 clips for `wav2vec-2.0`. The results are split into non-overlapping clips OVR $= 0.00$ and overlapping clips OVR > 0.00.

For each clip-model pair, the difference is measured as

$$d_i = \text{DSS-WER}_i - \text{WER}_i. \tag{5.10}$$

Negative values of d_i mean that DSS-WER gives a lower error estimate than WER. Following the experimental design in Section 4.9, the permutation tests use 10,000 random sign flips of the paired differences. The non-overlap test is two-sided because DSS-WER should ideally match WER exactly when there is no interleaving. The overlap test is one-sided in the lower-tail direction because the expected effect is that DSS-WER should be less than WER when there is interleaving. The confidence intervals are 95% bootstrap intervals for the mean paired difference, based on 10,000 resamples.

Table 5.6: DSS-WER versus WER on *non-overlapping* synthetic clips. Differences are DSS-WER minus WER. The p-values are from two-sided Monte Carlo paired sign-flip tests with 10,000 permutations.

Model	n	**WER mean (SD)**	**DSS mean (SD)**	**Mean diff. [95% CI]**	p
faster-whisper	100	0.052 (0.065)	0.061 (0.087)	+0.0089 [0.0036, 0.0155]	< 0.001
WhisperX	100	0.038 (0.039)	0.039 (0.040)	+0.0007 [0.0002, 0.0016]	0.015
wav2vec-2.0	1200	0.246 (0.317)	0.245 (0.313)	-0.0002 [-0.0009, 0.0004]	0.519
Parakeet	100	0.307 (0.232)	0.316 (0.243)	+0.0094 [0.0045, 0.0156]	< 0.001
All models	1500	0.223 (0.299)	0.224 (0.297)	+0.0011 [0.0004, 0.0019]	0.005

Table 5.7: DSS-WER versus WER on overlapping synthetic clips. Differences are DSS-WER minus WER. The p-values are from one-sided lower-tail Monte Carlo paired sign-flip tests with 10,000 permutations.

Model	n	**WER mean (SD)**	**DSS mean (SD)**	**Mean diff. [95% CI]**	p
faster-whisper	600	0.176 (0.128)	0.172 (0.123)	-0.0049 [-0.0075, -0.0023]	< 0.001
WhisperX	600	0.205 (0.148)	0.202 (0.149)	-0.0030 [-0.0052, -0.0009]	0.002
wav2vec-2.0	3600	0.433 (0.241)	0.405 (0.241)	-0.0277 [-0.0287, -0.0267]	< 0.001
Parakeet	600	0.536 (0.184)	0.548 (0.195)	+0.0116 [0.0086, 0.0147]	1.000
All models	5400	0.390 (0.243)	0.372 (0.243)	-0.0180 [-0.0190, -0.0171]	< 0.001

Table 5.8: Direction of the paired DSS-WER and WER differences. Percentages are within each model and overlap subset.

Subset	**Model**	**DSS = WER**	**DSS < WER**	**DSS > WER**
Non-overlap	faster-whisper	81 (81.0%)	2 (2.0%)	17 (17.0%)
	WhisperX	93 (93.0%)	0 (0.0%)	7 (7.0%)
	wav2vec-2.0	759 (63.3%)	197 (16.4%)	244 (20.3%)
	Parakeet	72 (72.0%)	4 (4.0%)	24 (24.0%)
	All models	1005 (67.0%)	203 (13.5%)	292 (19.5%)
Overlap	faster-whisper	279 (46.5%)	215 (35.8%)	106 (17.7%)
	WhisperX	365 (60.8%)	152 (25.3%)	83 (13.8%)
	wav2vec-2.0	312 (8.7%)	2999 (83.3%)	289 (8.0%)
	Parakeet	246 (41.0%)	166 (27.7%)	188 (31.3%)
	All models	1202 (22.3%)	3532 (65.4%)	666 (12.3%)

Table 5.6 shows that DSS-WER and WER are very close in the non-overlapping condition, with mean paired differences of 0.0011. However, the differences are statistically significant overall, as it contains 1500 pairs. That means even though the control condition doesn't show a large mean difference, the approximation is not perfect and still overestimates the error by a

small amount on average. The model exception is `wav2vec-2.0`, where the confidence interval includes zero, and the p-value is not significant ($p = 0.519$), which suggests that DSS-WER is a very close approximation to WER. This suggests that the DSS-WER algorithm is not just simply giving a lower error estimate but is actually able to find the correct path when there is no interleaving.

Table 5.7 shows the advantage of DSS-WER over WER in the overlapping condition, with mean paired differences of -0.0180 overall, a confidence interval that does not include zero ($[-0.0190, -0.0171]$) and a p-value of < 0.001. The strongest result is for `wav2vec-2.0`, where the mean paired difference is -0.0277 with a confidence interval of $[-0.0287, -0.0267]$ and a p-value of < 0.001 while `faster-whisper` and `WhisperX` also show a statistically significant reduction. This suggests that DSS-WER can significantly reduce the overestimation of error under overlapping speech for `wav2vec-2.0`. However, for `Parakeet`, the mean paired difference is positive at $+0.0116$ with a confidence interval of $[0.0086, 0.0147]$ and a p-value of 1.000, which indicates that DSS-WER actually gives a higher estimate than WER on average. This might be because `Parakeet` performs much worse than the other models, so the beam search algorithm of DSS-WER might be more likely to miss the correct path in the search space. In the case of a correct hypothesis, there are paths with significantly lower error, so the algorithm can easily narrow down to a few paths. On the other hand, if the hypothesis completely missed the reference, there might be many equally bad paths. Since the algorithm has a fixed beam width, it might not be able to explore all paths and thus miss the correct path, leading to a higher error rate. This suggests that the advantage of DSS-WER over WER under overlapping might depend on the hypothesis's accuracy.

Table 5.8 gives the clip-level direction of the paired differences. In the non-overlapping condition, about 67% of pairs have DSS-WER equal to WER, supporting the use of non-overlapping clips as a control. In the overlapping condition, about 88% of the pairs have DSS-WER less than or equal to WER, which supports the expected advantage of DSS-WER under overlapping speech. However, it also suggests that 12% of the pairs have DSS-WER greater than WER, which should be kept in mind when interpreting the results.

Overall, the results suggest that DSS-WER gives a statistically significant but small overestimation of error in the non-overlapping condition, and a statistically significant reduction in error in the overlapping condition, especially for `wav2vec-2.0`. In further applications of DSS-WER, it is important to keep in mind that the advantage of DSS-WER under overlapping might depend on the hypothesis's accuracy. For future use, it might be worth revisiting WER if there is no overlap. On the other hand, to improve the algorithm, it might be worth considering doing some hyperparameter tuning of the beam search algorithm, such as increasing beam width or lookahead steps or changing the heuristic, to see if it can further reduce the error under overlapping speech.

5.10 Summary of Findings

This chapter has presented the results of the statistical analysis for all 4 research questions. For RQ1, the results show that WER increases with higher OVR and lower SNR, which is consistent with the expected trends. For RQ2, the results show that `faster-whisper` has the best WER but the worst inference time, while `wav2vec-2.0` has the best inference time but a higher WER, and `WhisperX` is a balance between the two. For RQ3, the results show that the synthetic ranking transfers only partially to real data, with `Parakeet` performing best among the four ASR models. For RQ4, the results show that DSS-WER provides a statistically significant advantage over WER under the overlapping condition but shows a small overestimation in the non-overlapping condition.

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Appendix A

Pseudocode for DSS-WER Beam Search

Algorithm 1: DSS-WER-BEAM-2CHAIN(A, B, H)

Input: Reference word chains A, B and hypothesis H
Output: Approximate overlap-aware edit distance and WER

```
initialise search parameters B, λ, M, L;
R_front ← BeamPass(A, B, H, front);
R_tail ← BeamPass(A, B, H, tail);
return the better of R_front and R_tail under lexicographic order (distance, wer)
```

Algorithm 2: BEAMPASS$(A, B, H, \textit{direction})$

Input: Reference chains A, B, hypothesis H, and search direction
Output: One-pass approximate edit distance and WER

```
if direction = front then
    (Â, B̂, Ĥ) ← (A, B, H);
else
    (Â, B̂, Ĥ) ← (reverse(A), reverse(B), reverse(H));
normalise Â, B̂, Ĥ if required;
initialise bestCost[(0,0,0)] ← 0, backpointers, and beam ← {(0,0,0)};
while beam is non-empty and expansion budget not exhausted do
    generate all valid successors of each beam state using: match/substitute against A,
      match/substitute against B, insert, delete from A, and delete from B;
    keep only the best predecessor for each successor state;
    score each successor by
        f(i, j, t) = g(i, j, t) + λ h(i, j, t),                    (A.1)
      where g is accumulated edit cost and h is a lower-bound heuristic based on
      remaining-length mismatch and local bag-of-words mismatch;
    sort candidates by score and prune to the next beam using fixed or adaptive beam
      width, optionally with stratified selection;
read the final state (|Â|, |B̂|, |Ĥ|);
compute WER = distance/(|A| + |B|) when finite;
optionally reconstruct alignment from backpointers;
if direction = tail then
    reverse the recovered alignment;
return distance, WER, and alignment
```

Appendix B

Software Configuration

B.1 Software Dependencies

- numpy 2.4.4
- pandas 3.0.2
- soundfile 0.13.1
- librosa 0.11.0
- faster-whisper 1.2.1
- transformers 4.57.6
- torch 2.8.0
- whisperx 3.8.5

B.2 ASR Models and Configurations

faster-whisper large-v3 model with compute type int8

wav2vec-2.0 wav2vec2-large-960h

parakeet parakeet-tdt-0.6b-v3 with configuration: `right_context_secs=2.0`, `left_context_secs=10.0`, `chunk_secs=10`, `batch_size=1`, decoding=greedy

whisperx large-v2 model with compute type int8